

The pair singular series of a polynomial IV: the second spectrum

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Working draft of 13 September 2026 (corrected twice; see the notice)

Abstract

Part III reduced the Cesàro form of the off-diagonal of the pair singular series of an irreducible quadratic f to “pieces” $\mathcal{P}_u(Y)$, one for each way the two roots of f can agree on a part u of the modulus, and showed that everything except the pieces’ “windows” is understood. This paper is about what the pieces actually are. Numerically each piece is $O(\sqrt{Y})$, and $\mathcal{P}_u(Y)/\sqrt{Y}$ oscillates in $\log Y$ at the Laplace eigenvalues of the *even* Maass cusp forms of $\mathrm{SL}_2(\mathbb{Z})$, with amplitudes governed by the values of the forms at the Heegner points, or their integrals over the closed geodesics, of discriminant D : a second spectrum, next to the zeros of ζ_K that govern the diagonal (part II). We explain this by an explicit formula: the piece is a twisted Walfisz sum whose Dirichlet series is a sum over frequencies of Dirichlet series of Salié sums, which the Kuznetsov formula of half-integral weight continues to the half-line with poles at $\frac{1}{2} \pm it_j$, and the Katok–Sarnak correspondence identifies the residues as periods of Maass forms. The same law, and the same parameter-free prediction of its phases, holds for a barer and more classical object: the partial sums of the Weyl sums of the congruence $b^2 \equiv D$, which Hooley bounded by $X^{3/4+\varepsilon}$ and which we find to exhibit square-root cancellation with an explicit spectral expansion. For $u = 1$ we derive it on a model object, proving the exact spectral structure and the main terms but not yet the bound on the remainder (Remark 26); for smooth windows in place of the sharp cut-off the same computation gives a complete theorem (Theorem 27), with an absolutely convergent spectral expansion whose parameter-free phase prediction the data confirm to 0.035 radians. The route: the root pairs are Heegner points, the piece is a sum of one test function over a modular orbit, the convergence obstruction and the passage to L^2 are resolved by a truncation and by the symmetry of the orbit, the constant and continuous contributions vanish or are negligible because the seed has vanishing horizontal mean, the arising Bessel integral is evaluated exactly by a Mellin–Barnes contour shift, and the spectral sum converges absolutely for Riesz means of order at least two. The argument runs at the levels $\Gamma_0(e)$, $e \mid \mathrm{rad}(2D)$: the coprimality condition $(d, 2D) = 1$ carried by the moduli is not invariant under $\mathrm{SL}_2(\mathbb{Z})$, an error of the first version of this paper found by an external reader, but its Möbius resolution into the sub-families $\{e \mid d\}$ is invariant under $\Gamma_0(e)$, and the spectrum accordingly contains the even Maass newforms of those levels next to the level-one forms, as the data confirm (for $D = -3$ the largest line is the first even newform of level 3). We discuss the uniformity in u that the Cesàro conjecture of part I would need, and report the numerical evidence, including phase predictions with no free parameter that agree to a median of 0.012 radians.

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Correction notice (13 September 2026). An external assessment of the repository (revision a372a63, 12 September 2026; file `research/reviews/`) found that the proof of Theorem 25 in the version of 12 September replaced the restricted sum defining the model object (25), whose moduli satisfy $(d, 2D) = 1$, by a sum over complete $\mathrm{SL}_2(\mathbb{Z})$ -orbits of Heegner points, and that the restriction is not orbit-invariant: for $D = -4$ the admitted form $[5, 4, 1]$ and the excluded form $[2, 2, 1]$ lie in the same orbit. We verified this; the theorem was not established as then stated. The present version repairs the argument by working at the levels $\Gamma_0(e)$, $e \mid \mathrm{rad}(2D)$ (Lemma 11 and Step 2 of §5), which also required moving the parametrisation from discriminant D to discriminant $4D$ (Step 1); every analytic step is unchanged, the level being fixed by D . The repaired theorem predicts the even Maass newforms of the levels e in the spectrum of the model object and their absence from the unrestricted divisor sum, and this was checked numerically before the repair was written (Remark 13). The repair was then read adversarially by a fresh reader (13 September, later): the repaired Steps 1–3 and Lemma 11 passed, but four defects were found in the *analytic* bookkeeping of the original proof, which the assessment had not examined — the truncated-tail bound discards a factor $e^{-\pi t_j/2}$ that the normalisation of the coefficients gives back, so it does not sum over j ; the non-oscillatory error terms receive no Riesz decay in j ; the boundedness of the Poincaré series was asserted from a measure statement; a paragraph claimed a saving where the arithmetic gives a loss. The third is repaired (Lemma 18) and the fourth corrected. For the first two, the route of smoothing first was carried out the same evening (Proposition 24): it gives the exact spectral coefficients and the main terms, and it shows that the remainder is not a technicality — it contains Riesz-smoothed Hooley sums of the same order as the main terms, so that the coefficients of the theorem are the residues of Proposition 24 plus a part not yet computed (Remark 26). Theorem 25 stands as an exact expansion with proved leading terms; the bound on its remainder is open. What is proved in full is the smooth-window version, Theorem 27 (added the same night, tested before it was written). Record: `research/ERRATA.md`, items 13, 16, 17 and 19.

1 Introduction

In plain terms. Parts I and II showed that the main part of the fluctuation of prime values of a polynomial is governed by the zeros of a Dedekind zeta function. The remaining part, the “off-diagonal” of part III, turns out to be governed by a second, entirely different spectrum: the Laplace eigenvalues of the modular surface, the same numbers that govern the distribution of the roots of $x^2 \equiv D \pmod{d}$ (Hooley, Bykovskii, Duke–Friedlander–Iwaniec). Only half of them appear, those of the even forms, and the size of each contribution is the value of the corresponding eigenfunction at a special point, or its integral along a special closed curve, determined by the discriminant of the polynomial. We found this in the data before we had the theory; the theory, once written down, predicts exactly the pattern of presence and absence that the data show.

Set-up. Notation as in part III: $f = t^2 + bt + c$, $D = b^2 - 4c$, $\lambda(p) = p/(p - 2\omega(p))$; for squarefree u with $\omega(u) \geq 1$ and $Q_u(x) = u^2x^2 - D$,

$$\mathcal{P}_u(Y) = \sum_{h \leq Y} (Y - h) (F_u(Q_u(h)) - \mathbb{E}F_u) - \frac{1}{2}(\mathbb{E}F_u - 1)Y,$$

$$F_u(n) = \sum_{\substack{d \mid n, \, d > 1 \\ d \text{ admissible}}} \frac{\lambda(d)}{d}, \quad \mathbb{E}F_u = \sum_{\substack{d > 1 \\ d \text{ adm}}} \frac{\lambda(d)\omega(d)}{d^2},$$

admissible meaning squarefree, all primes split, coprime to $2Du$. Part III, Theorem A': $\mathrm{Off}_f^*(H) = c_{\mathrm{off}}(f)H + \sum_{u \leq H^{2/3+\varepsilon}} w(u)\mathcal{W}_u(H/u; \log H) + O(H(\log H)^{1-c})$, and the Cesàro form of Conjecture 1 of part I is equivalent to the windows being $o(H \log H)$.

Results.

- *The spectral formula for a piece* (Theorem 25; the exact expansion and its main terms are proved, the bound on the remainder is not — see the correction notice and Remark 26): for $u = 1$, fixed f , Riesz means of order $m \geq 2$, and the model object (25) of §6 rather than the piece itself (the weight λ and the squarefree condition are dropped there, the condition $(d, 2D) = 1$ is kept),

$$Y^{-m} \mathcal{P}_u^{(m)}(Y) = c_{u,m} Y + Y^{1/2} \sum_j (\alpha_j(u, f) Y^{it_j} + \overline{\alpha_j(u, f)} Y^{-it_j}) \gamma_m(t_j) + O(Y^{1/2-\delta}),$$

the sum over the even Maass cusp forms u_j of level one and the even newforms of the levels $e \mid \text{rad}(2D)$, with Laplace eigenvalue $\frac{1}{4} + t_j^2$ (and, for $u > 1$, the levels dividing $4u^2$), with $\alpha_j(u, f)$ an explicit signed combination of periods of u_j over Heegner points of level e and discriminant $4D$, the Katok–Sarnak periods for level one, and $\gamma_m(t) \ll |t|^{-m-1}$.

- *The smooth-window theorem* (Theorem 27, proved): for $w \in C_c^\infty((0, \infty))$ the smooth-window sum $\sum_h w(h/Y) (\sigma_{-1}^*(h^2 - D) - \mathbb{E} \sigma_{-1}^*)$ equals $Y^{-1/2}$ times an absolutely convergent spectral sum over the even Maass cusp forms of the levels $e \mid \text{rad}(2D)$, with explicit coefficients (periods over the sub-families, Fourier coefficients, $\Gamma(-it_j)$, the Mellin transform of w), a continuous part $o(Y^{-1/2})$ and an error $O(Y^{-5/2+\varepsilon})$; the phase of the first line is predicted with no free parameter and observed to 0.035 radians.
- *Numerical confirmation* (§6): for five quadratics the even parameters explain the variance of $\mathcal{P}_u(Y)/\sqrt{Y}$ at the 92nd–100th percentile among random frequency sets, the odd ones at the 12th–66th; the dominant line $t_1 = 13.7798$ has stable amplitude and phase over two decades of $\log Y$; the two polynomials with exponentially small periods ($D = -8, -163$) show nothing.
- *What the Cesàro conjecture needs* (§7): the sum of the pieces over $u \leq H^{2/3}$ converges to $c_{\text{off}} H + o(H)$ provided the spectral formula holds with an error $O((H/u)^{1/2+\varepsilon} u^A)$, $A < \frac{1}{4}$; the level dependence of the Kuznetsov formula is therefore the whole remaining problem.

2 The piece as a twisted Walfisz sum

Throughout, f is an irreducible quadratic of discriminant D and u is squarefree with $\omega(u) \geq 1$ prime factors. Call a modulus d *admissible* (for u) if it is squarefree, coprime to $2Du$, and every prime dividing it is split, $(\frac{D}{p}) = 1$, as in part III. The third condition is automatic for the moduli that actually divide a value: if $p \mid Q_u(h) = u^2 h^2 - D$ and $p \nmid 2Du$ then $D \equiv (uh)^2 \pmod{p}$, so p is split. It is nevertheless imposed, because the sums over admissible d below (the expectation $\mathbb{E} F_u$, the sawtooth series) run over all admissible moduli and not only over divisors of values, and for an inert prime p the root set R_p is empty and $\lambda(p)$ is not defined. For admissible d put

$$R_d = R_d^{(u)} = \{x \bmod d : u^2 x^2 \equiv D\}, \quad \rho(d) = |R_d| = 2^{\nu(d)}, \quad \rho_k(d) = \sum_{x \in R_d} e\left(\frac{kx}{d}\right),$$

ν the number of prime factors and $e(y) = e^{2\pi i y}$. Two properties are used constantly: R_d is *symmetric*, $x \in R_d \Rightarrow -x \in R_d$, and $0 \notin R_d$, both because $(D, d) = 1$; consequently every $\rho_k(d)$ is real. Write $\lambda(p) = p/(p-4)$ on split primes, λ multiplicative, and

$$F_u(n) = \sum_{\substack{d \mid n, \\ d > 1 \\ d \text{ admissible}}} \frac{\lambda(d)}{d}, \quad \mathbb{E} F_u = \sum_{\substack{d > 1 \\ d \text{ admissible}}} \frac{\lambda(d) \rho(d)}{d^2}, \quad Q_u(h) = u^2 h^2 - D, \quad (1)$$

both sums absolutely convergent ($\lambda(d)\rho(d) \ll d^\varepsilon$). All sums over moduli below run over admissible $d > 1$: including $d = 1$ would add 1 to F_u and to $\mathbb{E}F_u$ and would spoil the symmetry used in Lemma 1, since $R_1 = \{0\}$ is not a union of pairs $\{x, -x\}$ with $x \neq -x$. The *sharp piece* is the function of the real variable $t \geq 0$

$$S_u(t) = \sum_{h \leq t} F_u(Q_u(h)) - t \mathbb{E}F_u, \quad (2)$$

and the *Riesz pieces* are its Riesz means

$$\mathcal{P}_u^{(m)}(Y) = \frac{1}{m!} \int_0^Y (Y-t)^m S_u(t) dt \quad (m \geq 0), \quad (3)$$

the convention used throughout §§5–5.2. The expectation is subtracted at the real argument t , not at $\lfloor t \rfloor$; the two agree at integers, and the sawtooth identity of Lemma 1 holds for all real t only with this choice. For integer Y , since $\int_h^Y dt = Y - h$,

$$\begin{aligned} \mathcal{P}_u^{(0)}(Y) &= \sum_{h \leq Y} (Y-h)(F_u(Q_u(h)) - \mathbb{E}F_u) - \frac{1}{2} \mathbb{E}F_u Y \\ &= \mathcal{P}_u(Y) - \frac{Y}{2}, \end{aligned}$$

with $\mathcal{P}_u$ the piece of the set-up paragraph, the object computed in §6; the two differ by a linear term, which every statement below carries as a nuisance term anyway. The connection with part III is that $F_u(Q_u(h))$ is the divisor-sum form of the inner sum of a piece: $d \mid Q_u(h)$ exactly when h lies in one of the $\rho(d)$ classes R_d .

Lemma 1 (sawtooth form). *Let $\psi(y) = \frac{1}{2} - \{y\}$. For every $t \geq 0$,*

$$S_u(t) = \sum_{d \text{ admissible}} \frac{\lambda(d)}{d} \sum_{x \in R_d} \psi\left(\frac{t-x}{d}\right), \quad (4)$$

the series converging when the two terms $x, -x$ of each modulus are paired (it does not converge absolutely).

Proof. The right-hand side is not absolutely convergent term by term, since $\sum_d \lambda(d)\rho(d)/d$ diverges; it is summed with the terms x and $-x$ grouped, and converges in that sense, as shown at the end. Expanding F_u and counting,

$$\sum_{h \leq t} F_u(Q_u(h)) = \sum_d \frac{\lambda(d)}{d} \#\{h \leq t : d \mid Q_u(h)\} = \sum_d \frac{\lambda(d)}{d} \sum_{x \in R_d} \#\{h \leq t : h \equiv x \pmod{d}\},$$

the interchange being legitimate because $\sum_{d \mid Q_u(h)} \lambda(d)/d$ converges absolutely for each $h \leq t$ and there are finitely many h . Represent each $x \in R_d$ by $\langle x \rangle_d \in [1, d]$. For $0 \leq t$ and $1 \leq x \leq d$,

$$\#\{1 \leq h \leq t : h \equiv x \pmod{d}\} = \left\lfloor \frac{t-x}{d} \right\rfloor + 1,$$

valid also when $x > t$, where both sides vanish because $-1 < (t-x)/d < 0$. Writing $\lfloor y \rfloor = y - \frac{1}{2} + \psi(y)$,

$$\#\{h \leq t : h \equiv x \pmod{d}\} = \frac{t}{d} + \left(\frac{1}{2} - \frac{x}{d}\right) + \psi\left(\frac{t-x}{d}\right).$$

Now sum over $x \in R_d$. Since R_d is symmetric and $0 \notin R_d$, its representatives split into pairs $\{\langle x \rangle_d, d - \langle x \rangle_d\}$, and $(\frac{1}{2} - \frac{x}{d}) + (\frac{1}{2} - \frac{d-x}{d}) = 0$; hence $\sum_{x \in R_d} (\frac{1}{2} - \langle x \rangle_d/d) = 0$ for every admissible $d > 1$. Therefore

$$\sum_{h \leq t} F_u(Q_u(h)) = t \sum_d \frac{\lambda(d)\rho(d)}{d^2} + \sum_d \frac{\lambda(d)}{d} \sum_{x \in R_d} \psi\left(\frac{t-x}{d}\right) = t \mathbb{E}F_u + \sum_d \frac{\lambda(d)}{d} \sum_{x \in R_d} \psi\left(\frac{t-x}{d}\right),$$

which is (4). For the convergence, group the terms of a pair $\{x, -x\}$ and write $a = \langle x \rangle_d$ with $a \leq d/2$, so that the pair is $\{a, d - a\}$. If $d > 2t$ and $a > t$ then $\{(t - a)/d\} = (t - a + d)/d$ and $\{(t + a)/d\} = (t + a)/d$, so by the periodicity of ψ the two sawtooths sum to $-2t/d$. The alternative $a \leq t$ means that some $h \leq t$ satisfies $d \mid Q_u(h)$, hence $d \leq u^2 t^2 + |D|$. So for $d > D_0(t) := u^2 t^2 + |D|$, which exceeds $2t$, every pair contributes exactly $-2t/d$, and the tail of the series is

$$\sum_{d > D_0} \frac{\lambda(d)}{d} \cdot \frac{\rho(d)}{2} \cdot \left(-\frac{2t}{d}\right) = -t \sum_{d > D_0} \frac{\lambda(d)\rho(d)}{d^2} \ll t^{-1+\varepsilon},$$

an absolutely convergent sum, while the range $d \leq D_0$ is a finite sum. So the paired series converges, and (4) holds, for every real $t \geq 0$. $\square$

In words. A piece counts, for each admissible modulus d , how far the number of $h \leq t$ with $d \mid Q_u(h)$ departs from its expectation, weighted by $\lambda(d)/d$; and that departure is a sawtooth evaluated at the roots. The linear term that normally accompanies such a count cancels because the roots come in pairs $\pm x$. This is the same cancellation that makes the diagonal of part I a sum of sawtooths $\{H/d\} - \frac{1}{2}$, and it is why the piece is a Walfisz-type sum: in the classical case $\sum_{n \leq x} n^{-1}(\frac{1}{2} - \{x/n\})$ one sums a sawtooth over all moduli with weight $1/n$, and here one sums it over admissible moduli, at the roots of a quadratic congruence, with weight $\lambda(d)/d$.

Lemma 2 (Weyl-sum form). *For every $t \geq 0$,*

$$S_u(t) - \frac{1}{2}F_u(Q_u(t)) \mathbf{1}_{t \in \mathbb{Z}} = \frac{1}{\pi} \sum_{k \geq 1} \frac{1}{k} \sum_{d \text{ admissible}} \frac{\lambda(d)\rho_k(d)}{d} \sin \frac{2\pi kt}{d}, \quad (5)$$

in the following sense: for each D_0 the k -series over the finitely many moduli $d \leq D_0$ converges, and the truncated right-hand side tends to the left-hand side as $D_0 \rightarrow \infty$; for $D_0 \geq u^2 t^2 + |D|$ the omitted moduli contribute exactly $-t \sum_{d > D_0} \lambda(d)\rho(d)d^{-2}$. Consequently, for $m \geq 1$ and in the same sense,

$$\mathcal{P}_u^{(m)}(Y) = \frac{1}{\pi} \sum_{k \geq 1} \frac{1}{k} \sum_d \frac{\lambda(d)\rho_k(d)}{d} \Phi_m\left(\frac{2\pi k}{d}; Y\right), \quad \Phi_m(\theta; Y) = \frac{1}{m!} \int_0^Y (Y-t)^m \sin(\theta t) dt, \quad (6)$$

with $\Phi_m(\theta; Y) \ll_m \min(\theta Y^{m+2}, Y^m \theta^{-1})$.

Proof. The Fourier series $\sum_{k \geq 1} \sin(2\pi ky)/(\pi k)$ has uniformly bounded partial sums; it converges to $\psi(y)$ for $y \notin \mathbb{Z}$ and to $0 = \psi(y) - \frac{1}{2}$ for $y \in \mathbb{Z}$. Insert it into (4) for the finitely many $d \leq D_0$. For such d and $x \in R_d$ the argument $(t - x)/d$ is an integer exactly when $t \in \mathbb{Z}$ and $d \mid Q_u(t)$; these terms are the ones for which the series returns $\psi - \frac{1}{2}$ instead of ψ , and they account for the correction $\frac{1}{2} \sum_{d \mid Q_u(t)} \lambda(d)/d = \frac{1}{2} F_u(Q_u(t))$ on the left. For each d ,

$$\sum_{x \in R_d} \sin \frac{2\pi k(t-x)}{d} = \Im \left[e\left(\frac{kt}{d}\right) \overline{\rho_k(d)} \right] = \rho_k(d) \sin \frac{2\pi kt}{d},$$

since $\rho_k(d)$ is real. The tail $d > D_0$ is the one computed in the proof of Lemma 1. For (6) apply $\frac{1}{m!} \int_0^Y (Y-t)^m dt$ to (5): the correction term is supported on the integers and does not affect the integral; for each fixed $d \leq D_0$ the partial sums in k are uniformly bounded and converge for almost every t , so the integral may be taken termwise (dominated convergence); and the integrated tail, $-\frac{1}{m!} \int_0^Y (Y-t)^m t dt \sum_{d > D_0} \lambda(d)\rho(d)d^{-2}$, tends to 0 with D_0 . The bound on Φ_m follows from $|\sin \theta t| \leq \theta t$ and from one integration by parts. Neither (5) nor (6) is used in §§5–5.2, which work with the sawtooth form directly; they are recorded because they are the form in which the modulus-side route of §3 sees the problem. $\square$

¹The alternative $a \leq t$ does occur for $d > 2t$ ($D = -4$, $u = 1$, $t = 2$, $d = 5$, $a = 1$), so the finiteness of that range comes from $d \mid Q_u(a)$ and not from $d > 2t$; the first version argued the latter.

Remark 3. Two features of (6) decide everything that follows. First, the frequencies k enter only through the *Weyl sums* $\rho_k(d)$ of the dilated roots, so the arithmetic of the problem is entirely contained in the Dirichlet series

$$W_k(s) = \sum_{d \text{ admissible}} \frac{\lambda(d)\rho_k(d)}{d^s} \quad (\Re s > 1), \quad (7)$$

one for each k . Second, the weight $1/k$ in front decays only like k^{-1} , so a bound for W_k that loses a power of k is expensive: this is the same phenomenon that forced the condition $\theta + 6B < 1$ in part III, Theorem A, and it is the reason we do not bound the $\rho_k(d)$ one frequency at a time. The route of this paper is instead to continue $W_k(s)$ analytically, where the dependence on k appears as a Fourier coefficient of an automorphic form and therefore *decays* on average.

Proposition 4 (the Mellin representation). *Let $m \geq 1$ and let*

$$A_u(s) = \sum_{h \geq 1} \frac{F_u(Q_u(h)) - \mathbb{E}F_u}{h^s} \quad (\Re s > 1).$$

Then A_u extends to a function holomorphic in $\Re s > \frac{1}{2}$ except possibly for the singularities inherited from the W_k , and for integer $Y \geq 2$ and any $c > 1$,

$$\begin{aligned} \mathcal{P}_u^{(m)}(Y) &= \frac{1}{2\pi i} \int_{(c)} A_u(s) \frac{Y^{s+m+1}}{s(s+1) \cdots (s+m+1)} ds + \mathbb{E}F_u R_m(Y), \\ R_m(Y) &= \sum_{h \leq Y} \frac{(Y-h)^{m+1}}{(m+1)!} - \frac{Y^{m+2}}{(m+2)!}, \end{aligned} \quad (8)$$

R_m an explicit polynomial of degree $m+1$ in the integer Y (Faulhaber): by (3), $\mathcal{P}_u^{(m)}(Y) = \sum_{h \leq Y} \frac{(Y-h)^{m+1}}{(m+1)!} F_u(Q_u(h)) - \mathbb{E}F_u \frac{Y^{m+2}}{(m+2)!}$, and the integral is Perron's formula for the Riesz sum of order $m+1$ of the coefficients of A_u . Moreover, for $0 < \Re s < 1$,

$$A_u(s) = \frac{2\Gamma(1-s)}{(2\pi)^{1-s}} \sin \frac{\pi s}{2} \sum_{k \geq 1} k^{s-1} W_k(s+1) + (\text{entire}), \quad (9)$$

whenever the k -sum converges, where W_k is as in (7).

Proof. (8) is the Riesz–Perron formula, valid because $F_u(Q_u(h)) - \mathbb{E}F_u \ll h^\varepsilon$ so that A_u converges absolutely for $\Re s > 1$. For (9), expand F_u and sum over each residue class:

$$\sum_{h \geq 1} \frac{F_u(Q_u(h))}{h^s} = \sum_d \frac{\lambda(d)}{d^{1+s}} \sum_{x \in R_d} \zeta\left(s, \frac{\langle x \rangle_d}{d}\right),$$

$\zeta(s, \alpha)$ the Hurwitz zeta function. Since $\zeta(s, \alpha)$ has residue 1 at $s = 1$ for every α , the pole of the d -th term is $\rho(d)\lambda(d)d^{-1-s}$, and summing over d these residues total $\mathbb{E}F_u$, which is exactly the pole of $\mathbb{E}F_u \zeta(s)$: each bracket $\lambda(d)d^{-1-s} \sum_x \zeta(s, \langle x \rangle_d/d) - \lambda(d)\rho(d)d^{-2}\zeta(s)$ is entire, which is the statement that A_u continues past $s = 1$. Hurwitz's formula,

$$\zeta(s, \alpha) = \frac{2\Gamma(1-s)}{(2\pi)^{1-s}} \left[\sin \frac{\pi s}{2} \sum_{k \geq 1} \frac{\cos 2\pi k \alpha}{k^{1-s}} + \cos \frac{\pi s}{2} \sum_{k \geq 1} \frac{\sin 2\pi k \alpha}{k^{1-s}} \right],$$

valid for $\Re s < 0$ and by continuation for $\Re s < 1$ with the series interpreted as the analytic continuations of the periodic zeta functions, gives after summing over the symmetric set R_d

$$\sum_{x \in R_d} \zeta\left(s, \frac{\langle x \rangle_d}{d}\right) = \frac{2\Gamma(1-s)}{(2\pi)^{1-s}} \sin \frac{\pi s}{2} \sum_{k \geq 1} \frac{\rho_k(d)}{k^{1-s}},$$

the sine term vanishing identically because $\sum_{x \in R_d} \sin(2\pi k x/d) = 0$ by symmetry. Multiplying by $\lambda(d)d^{-1-s}$ and summing over d gives (9). $\square$

In words. The piece is determined by one Dirichlet series for each frequency, namely the Dirichlet series of the Weyl sums of the roots of $u^2x^2 \equiv D$; the symmetry of the root set kills the odd half of Hurwitz's formula, so only the even half survives. That the odd half disappears here is the first appearance of the parity phenomenon that will select the even Maass forms in §4, and it has the same source: the root set of a quadratic congruence is invariant under $x \mapsto -x$.

3 Salié sums and the Dirichlet series of the Weyl sums

By Remark 3 everything now depends on the series $W_k(s) = \sum_d \lambda(d) \rho_k(d) d^{-s}$. This section identifies its coefficients as Kloosterman sums of half-integral weight, which is what makes an analytic continuation possible. Write, for $c \geq 1$ odd with $(D, c) = 1$ and $h \in \mathbb{Z}$,

$$\mathcal{W}_h(D; c) = \sum_{b \bmod c, b^2 \equiv D} e\left(\frac{hb}{c}\right), \quad (10)$$

the Weyl sum of the roots of the *fixed* quadratic congruence, and let $K_\chi(m, n; p) = \sum_{x \bmod p}^* \left(\frac{x}{p}\right) e\left(\frac{mx+n\bar{x}}{p}\right)$ be the Kloosterman sum twisted by the Legendre symbol, $\varepsilon_p = 1$ or i according as $p \equiv 1$ or $3 \pmod{4}$.

Lemma 5 (the dilation is a change of frequency). *For admissible d and $k \geq 1$, with $\bar{u}$ the inverse of u modulo d ,*

$$\rho_k(d) = \mathcal{W}_{k\bar{u}}(D; d).$$

Proof. $x \mapsto b = ux$ is a bijection of $R_d^{(u)} = \{x : u^2x^2 \equiv D\}$ onto $\{b : b^2 \equiv D\}$ because $(u, d) = 1$, and $kx \equiv k\bar{u}b \pmod{d}$. $\square$

So the dilation by u does not change the quadratic; it moves the frequency from k to $k\bar{u}$, a frequency that depends on the modulus. This is the exact point at which the pieces with $u > 1$ leave the range of the theorems of Duke, Friedlander and Iwaniec, which are stated for frequencies fixed independently of the modulus.

Lemma 6 (Salié). *Let p be an odd prime with $\left(\frac{D}{p}\right) = 1$ and let $h \in \mathbb{Z}$ with $p \nmid h$. Then*

$$\mathcal{W}_{2h}(D; p) = \frac{1}{\varepsilon_p \sqrt{p}} K_\chi(D, h^2; p). \quad (11)$$

For odd squarefree $c = c_1 c_2$ with $(c_1, c_2) = 1$ and $(D, c) = 1$,

$$\mathcal{W}_h(D; c) = \mathcal{W}_{h\bar{c}_2}(D; c_1) \mathcal{W}_{h\bar{c}_1}(D; c_2), \quad (12)$$

so (11) determines $\mathcal{W}_h(D; c)$ for every odd squarefree c coprime to $2D$.

Proof. (12) is the Chinese remainder theorem: $b \mapsto (b \bmod c_1, b \bmod c_2)$ is a bijection of the roots modulo c onto the pairs of roots, and $b/c \equiv b_1 \bar{c}_2 / c_1 + b_2 \bar{c}_1 / c_2 \pmod{1}$. For (11), substitute $b = \bar{h}y$ in (10), which is a bijection of the roots of $b^2 \equiv D$ onto the roots of $y^2 \equiv h^2 D$ and turns $e(2hb/p)$ into $e(2y/p)$; hence $\mathcal{W}_{2h}(D; p) = \sum_{y^2 \equiv h^2 D} e(2y/p)$. Salié's evaluation of the twisted Kloosterman sum, $K_\chi(m, n; p) = \varepsilon_p \sqrt{p} \left(\frac{m}{p}\right) \sum_{y^2 \equiv mn \pmod{p}} e\left(\frac{2y}{p}\right)$ [8, §4.5], applied with $m = D$, $n = h^2$ gives (11), the factor $\left(\frac{D}{p}\right)$ being 1 because p is split. ² $\square$

²Both identities were verified numerically for all split $p \leq 37$, all $D \in \{-4, -3, 5, 8, 12\}$ and $h \in \{1, 2, 3, 5\}$, and (12) for composite c ; the reference for Salié's evaluation should be pinned to a precise statement of the normalisation. The first version substituted $b = 2\bar{h}y$, which turns $e(2hb/p)$ into $e(4y/p)$ and does not give the displayed intermediate identity.

Remark 7 (what Lemma 6 does and does not give). Combining Lemmas 5 and 6, for even k and admissible d the coefficient of $W_k(s)$ is, up to the elementary factor $\varepsilon_d^{-1}d^{-1/2}$, a Kloosterman sum of half-integral weight with the two arguments D and $(k\bar{u}/2)^2$: the first fixed by the polynomial, the second carrying the frequency and the dilation. This is the shape to which the Kuznetsov formula of weight one half applies, and it is the machinery behind every known estimate for these Weyl sums *on average over the modulus* [6, 2, 3].

It is, however, *not* the route to the oscillations observed in §6, and it is worth saying why, because the distinction is easy to miss. Selberg’s Kloosterman zeta function is normalised as $Z_{m,n}(\sigma) = \sum_c S(m,n;c)c^{-2\sigma}$, with poles at $\sigma = s_j$, $\lambda_j = s_j(1 - s_j)$ [4, Theorem 1], and their Remark 3 covers exactly our case, weight $\frac{1}{2}$ on $\Gamma_0(4N)$. Tracing that normalisation through Lemma 6 and Proposition 4 places the singularities of A_u at $s = -\frac{1}{2} + 2it_j$ and predicts an oscillation of $\mathcal{P}_u(Y)/\sqrt{Y}$ at frequency $2t_j$. That prediction is false: fitted at the frequencies $2t_j$ the even set explains between 2.1% and 2.2% of the variance and sits at the 43rd to 69th percentile of random frequency sets drawn from the same band, against the 99.7th to 100th percentile at the frequencies t_j (§6; `scripts/piece-freq.ts`). The reason for the mismatch is that $Z_{m,n}$ sums over the *modulus*, whereas $\mathcal{P}_u$ sums over the *argument* h . The two are different analytic problems, and only the second produces the observed law.

The argument-side expansion. The route that does fit is Bykovskii’s [1]: write the sum over $h \leq Y$ as a finite combination of Poincaré series for $\mathrm{SL}_2(\mathbb{Z})$ and apply the spectral decomposition in weight 0 and level 1. There the parametrisation $\lambda_j = s_j(1 - s_j)$, $s_j = \frac{1}{2} + it_j$, produces terms $Y^{s_j} = \sqrt{Y}Y^{it_j}$ directly, exactly as the error term in the hyperbolic lattice-point problem does. This accounts at once for all four features of the data: the frequencies are the t_j themselves, the scale is $\sqrt{Y}$, the spectrum is that of weight 0 and of the levels $e \mid \mathrm{rad}(2D)$ with the level-one lines dominant (the newforms of level e were seen in the data only after the first version of this paper had been corrected, Remark 13), and the forms that can appear are those pairing non-trivially with the Heegner points or the closed geodesics, which by Proposition 9 and Corollary 12 are the even ones. What we prove is the following; the proof occupies §§5–5.2, and the precise form, with the order of the Riesz mean, is Theorem 25. (Its first version, at level one only, was wrong; see the correction notice and Remark 13.)

Theorem 8 (the spectral formula for a piece). *Let $u = 1$ and let $\mathcal{G}$ be the model object (25), smoothed by a Riesz mean of sufficiently high order. There are coefficients $\alpha_j(D)$, supported on the even Maass cusp forms of the groups $\Gamma_0(e)$, $e \mid \mathrm{rad}(2D)$ — the forms of level one and the newforms of the levels e — with $\lambda_j = \frac{1}{4} + t_j^2$ and proportional to periods of u_j over Heegner points of level e and discriminant $4D$ (the Katok–Sarnak periods when $e = 1$), such that*

$$\mathcal{G}(Y) = \sqrt{Y} \sum_j (\alpha_j(D)Y^{it_j} + \overline{\alpha_j(D)}Y^{-it_j}) + \mathcal{E}(Y) + O(Y^{1/2-\delta})$$

for some $\delta > 0$, the sum converging absolutely and $\mathcal{E}$ being the contribution of the continuous spectrum, which is $o(\sqrt{Y})$.

For fixed D this is in substance the content of Bykovskii’s spectral treatment of the divisor sums $\sum_{|n| \leq P} \sigma_{-s}(n^2 + D)$ [1], which is where an asymptotic of this shape was first obtained; what we require beyond it is the identification of the oscillating terms, their amplitudes and the error exponent. The proof given in §5.2 does not go through the Weyl-sum series W_k at all; it works with the test function directly, which is what makes it elementary. Lemmas 5 and 6 are retained because the identification of the Weyl sums as Salié sums is what any *quantitative* treatment of the remainder, or any attempt at the uniformity in u of §7, will have to use.

4 Katok–Sarnak and the parity rule

In the modulus-side route of §3 the Weyl sums are, by Lemma 6, Kloosterman sums of half-integral weight, and the Kuznetsov formula of weight $\frac{1}{2}$ would place the singularities of W_k at $s = \frac{1}{2} \pm 2it_j$ (Selberg’s normalisation $c^{-2\sigma}$ together with the factor $d^{-1/2}$ of (11)), with residues proportional to $\overline{\rho_j(|D|)}\rho_j((k/2)^2)$, ρ_j the Fourier coefficients of the weight- $\frac{1}{2}$ forms. That is the frequency $2t_j$ rejected in Remark 7; the route of §5 produces t_j . What survives from the modulus side is the arithmetic of the coefficients $\rho_j(|D|)$: two classical inputs turn them into periods of the Maass forms of $\mathrm{SL}_2(\mathbb{Z})$, and that is what decides which forms can appear.

Shimura and Katok–Sarnak. The Shimura correspondence sends the weight- $\frac{1}{2}$ forms of level 4 in Kohnen’s plus space to weight-0 Maass forms of level 1 with the same spectral parameter, and the image consists of the *even* forms; the odd Maass forms correspond to weight $\frac{3}{2}$. Katok and Sarnak [11] identify the Fourier coefficients of the weight- $\frac{1}{2}$ form at a fundamental discriminant D with a period of the corresponding Maass form u_j : for $D < 0$ the sum of u_j over the Heegner points of discriminant D , and for $D > 0$ the integral of u_j over the closed geodesics of discriminant D . Writing $\mathrm{Per}_D(u_j)$ for that period,

$$\rho_j(|D|) = \kappa(t_j, D) \mathrm{Per}_D(u_j) \quad (13)$$

with an explicit factor κ .³

Proposition 9 (parity). *If u_j is an odd Maass form then $\mathrm{Per}_D(u_j) = 0$, for every discriminant D . Consequently only the even Maass forms of $\mathrm{SL}_2(\mathbb{Z})$ can contribute to the pieces.*

Proof. Let $\iota(z) = -\bar{z}$, the reflection in the imaginary axis; u_j is odd exactly when $u_j \circ \iota = -u_j$. For $D < 0$ the Heegner points of discriminant D are the roots $z_Q \in \mathbb{H}$ of the integral binary quadratic forms $Q = [a, b, c]$ of discriminant D , one for each class, and $\iota(z_{[a,b,c]}) = z_{[a,-b,c]}$; since $[a, b, c] \mapsto [a, -b, c]$ is the inversion of the class group, it permutes the classes, so ι permutes the set of Heegner points. Hence $\mathrm{Per}_D(u_j) = \sum_Q u_j(z_Q) = \sum_Q u_j(\iota z_Q) = -\mathrm{Per}_D(u_j)$, which forces $\mathrm{Per}_D(u_j) = 0$. For $D > 0$ the same involution permutes the closed geodesics S_Q attached to the classes, reversing the orientation of each; the cycle integral $\int_{\Gamma_Q \backslash S_Q} u_j ds$ is taken against the unoriented hyperbolic arc length, so the orientation is immaterial and the same computation applies to $\mathrm{Per}_D(u_j) = \sum_Q \int_{\Gamma_Q \backslash S_Q} u_j ds$. $\square$

In words. The Heegner points of a given discriminant, and likewise the closed geodesics, are symmetric about the imaginary axis; an odd Maass form is antisymmetric about it; so its period vanishes. This is the third appearance of one and the same symmetry. In §2 it made the linear term of the sawtooth identity vanish; in Proposition 4 it killed the odd half of Hurwitz’s formula; here it removes the odd half of the spectrum. All three are the statement that the roots of a quadratic congruence come in pairs $\pm x$.

Corollary 10 (the shape of a piece). *In Theorem 8 the spectral sum runs over even Maass cusp forms only, those of $\mathrm{SL}_2(\mathbb{Z})$ and the newforms of the levels $e \mid \mathrm{rad}(2D)$. The coefficient of a line t_j is a signed combination, over e , of the periods of the corresponding forms over the sub-families $\mathcal{H}_e$ of Lemma 11 (Step 3 of §5); for the unrestricted divisor sum, which is the term $e = 1$ alone, it is proportional to the Katok–Sarnak periods $\sum_{g^2 \mid 4D} \mathrm{Per}_{4D/g^2}(u_j)$.*

³The identity is linear in the period, as in [11]; its square, which expresses $|\rho_j(|D|)|^2$ through $L(\frac{1}{2}, u_j \otimes \chi_D)$, is the Waldspurger–Kohnen–Zagier form and is not what is used here. In the level- N version announced by Sugiyama [16], the coefficient of the weight- $\frac{1}{2}$ form at $-n$ is $n^{-3/4} \sum_v \Phi(z_v)/|\mathrm{Stab}(v)|$, the sum over the $\Gamma_0(N)$ -classes of the forms of the corresponding discriminant; so κ carries the factor $|D|^{-3/4}$ and, in that normalisation, no dependence on t_j . Only $|\kappa|$, and only its dependence on D , enters below; the constant should still be pinned to [11] before any amplitude is computed from (13).

The size of the periods, and a visibility threshold. For $D < 0$ of class number one the period is the single value $u_1(z_D)$ at $z_D = (-b + i\sqrt{|D|})/2$. Through the Fourier expansion $u_j(z) = 2\sqrt{y} \sum_{n \geq 1} a_j(n) K_{it_j}(2\pi ny) \cos(2\pi nx)$ this is governed by $K_{it_j}(2\pi y)$ with $y = \sqrt{|D|}/2$. The Bessel function $K_{it}(x)$ of imaginary order is of size $e^{-\pi t/2}$ and oscillatory while $x < t$; for $x > t$ it decays like $\exp(-(\sqrt{x^2 - t^2} - t \arccos(t/x)))$, which is gradual just past the turning point, about $\exp(-\frac{2\sqrt{2}}{3}(x - t)^{3/2}t^{-1/2})$, and exponential only well beyond it. The turning point $x = t_j$ is

$$|D| = \left(\frac{t_j}{\pi}\right)^2, \quad (14)$$

about 19.2 for the first even form, $t_1 = 13.7798$. Explicitly, $|K_{it_1}(\pi\sqrt{|D|})|$ relative to its value at $|D| = 3$ is 1.0 at $|D| = 23$, 0.48 at 27, 0.023 at 43, $3 \cdot 10^{-4}$ at 67 and 10^{-9} at 163. Below the turning point the periods of a fixed form are of comparable size only on average: $K_{it_1}(\pi\sqrt{|D|})$ changes sign between $|D| = 3$ and $|D| = 4$, so a discriminant close to a zero has an anomalously small period. The falsifiable prediction is therefore: the first line is visible for the small discriminants, marginal around $|D| = 27$, reduced by a factor of order 50 at $|D| = 43$, and gone at 67 and 163; the higher lines, having larger t_j , persist further. Section 6 reports what is observed.

5 The spectral formula for a piece

We now give the argument-side route for $u = 1$ and the model object (25). The moduli of the model carry the condition $(d, 2D) = 1$, and the first version of this section (12 September 2026) treated the sum as a sum over complete $\mathrm{SL}_2(\mathbb{Z})$ -orbits of Heegner points, which it is not (Remark 13). The argument below works instead at the levels $e \mid \mathrm{rad}(2D)$: the condition is resolved by Möbius inversion into sub-families $\{e \mid d\}$, each a finite union of orbits of $\Gamma_0(e)$, and the spectral theory of $\Gamma_0(e) \backslash \mathbb{H}$ replaces that of the modular surface. Nothing analytic changes; the level is fixed, depending on D alone. Throughout $D < 0$; for $D > 0$ every occurrence of a Heegner point is replaced by a closed geodesic and the argument is parallel.

Step 1: the root pairs are Heegner points, of discriminant $4D$. A pair (d, x) with $d \geq 1$ and $x \bmod d$, $x^2 \equiv D \pmod{d}$, determines the integral form $[d, 2x, (x^2 - D)/d]$ of discriminant $4D$, and every form $[d, b, c]$ of discriminant $4D$ with $d \geq 1$ arises exactly once in this way: $b^2 - 4dc = 4D$ forces b even, $b = 2x$ with $x^2 \equiv D \pmod{d}$, and $b \bmod 2d$ corresponds to $x \bmod d$. This is Gauss's parametrisation in the form used by Hooley [6, §3] and studied in general by Welsh [17], and it holds for *every* d , even or odd, coprime to D or not; the forms need not be primitive. Writing $\mathcal{Q}_{4D}$ for the set of all integral forms $Q = [a, b, c]$ of discriminant $4D$ with $a \geq 1$, primitive or not, and $z_Q = (-b + \sqrt{4D})/(2a)$ for the root of Q in $\mathbb{H}$,

$$\{(d, x) : d \geq 1, x \bmod d, x^2 \equiv D \pmod{d}\} \longleftrightarrow \mathcal{Q}_{4D}, \quad (d, x) \mapsto z_{(d, x)} = \frac{-x + \sqrt{D}}{d}. \quad (15)$$

A form of content g is g times a primitive form of discriminant $4D/g^2$ with the same root, so the points $z_{(d, x)}$ are the Heegner points of the discriminants $4D/g^2$, $g^2 \mid 4D$, and they fall into $\sum_g h(4D/g^2)$ orbits of $\Gamma = \mathrm{PSL}_2(\mathbb{Z})$.⁴ The two coordinates of a point carry exactly the two data of a root pair:

$$\Im z_{(d, x)} = \frac{\sqrt{|D|}}{d}, \quad \Re z_{(d, x)} = -\frac{x}{d}. \quad (16)$$

⁴Verified numerically: reducing the forms $[d, 2x, *]$ for all $d \leq 80$ gives exactly $\sum_g h(4D/g^2)$ classes, namely 2, 2, 2, 3, 4, 4, 6, 6, 6 for $D = -4, -3, -7, -8, -11, -15, -20, -24, -23$. The first version of this paper parametrised the pairs by forms of discriminant D with $b^2 \equiv D \pmod{4d}$; that is a bijection onto the same pairs when d is odd and coprime to D (Lemma 17), but it is not available for the moduli the sub-families below contain, and it produces different points of $\mathbb{H}$.

Lemma 11 (the sub-families are $\Gamma_0(e)$ -stable). *For $e \geq 1$ let $\mathcal{H}_e = \{[a, b, c] \in \mathcal{Q}_{4D} : e \mid a\}$, the forms whose modulus is divisible by e . Then $\mathcal{H}_e$ is stable under $\Gamma_0(e)$ and under the reflection $\iota : [a, b, c] \mapsto [a, -b, c]$, and it is a finite union of $\Gamma_0(e)$ -orbits. For $e = 1$ it is all of $\mathcal{Q}_{4D}$.*

Proof. For $\gamma = \begin{pmatrix} \alpha & \beta \\ \gamma' & \delta \end{pmatrix} \in \Gamma_0(e)$ the leading coefficient of $Q \circ \gamma$ is $Q(\alpha, \gamma') = a\alpha^2 + b\alpha\gamma' + c\gamma'^2 \equiv a\alpha^2 \pmod{e}$, since $e \mid \gamma'$, and $(\alpha, e) = 1$ because $\alpha\delta - \beta\gamma' = 1$; so $e \mid a$ is preserved. The reflection fixes a . For the finiteness, $\mathcal{Q}_{4D}$ is a finite union of Γ -orbits, $\Gamma_0(e)$ has finite index in Γ , so each Γ -orbit is a finite union of $\Gamma_0(e)$ -orbits, and $\mathcal{H}_e$ is a union of some of them. ⁵ $\square$

Corollary 12 (parity at level e). *Let u be a Maass cusp form of $\Gamma_0(e)$ with $u \circ \iota = -u$, $\iota(z) = -\bar{z}$. Then $\sum_{Q \in \Gamma_0(e) \backslash \mathcal{H}_e} u(z_Q) / |\Gamma_0(e)_{z_Q}| = 0$, where $\Gamma_0(e)_{z_Q}$ is the stabiliser of z_Q in $\Gamma_0(e)$.*

Proof. ι normalises $\Gamma_0(e)$ (conjugation by ι negates the off-diagonal entries), so it permutes the $\Gamma_0(e)$ -orbits in $\mathcal{H}_e$, which it preserves by Lemma 11, and $\iota(z_Q) = z_{\iota Q}$ with stabilisers of equal order. The sum therefore equals its own negative. This is Proposition 9 at level e . $\square$

The moduli of (25) are those with $(d, 2D) = 1$, and $\mathbf{1}_{(d, 2D)=1} = \sum_{e \mid \text{rad}(2D)} \mu(e) \mathbf{1}_{e \mid d}$. The sub-families $\mathcal{H}_e$ with $e \mid \text{rad}(2D)$ are the objects the argument is about; there are $2^{\omega(2D)}$ of them. For odd $e \mid D$ the condition $e \mid a$ forces $e \mid b$ (as $x^2 \equiv D \equiv 0 \pmod{e}$ with e squarefree), so these are Heegner points of level e in the sense of Gross, Kohnen and Zagier, with root $b \equiv 0 \pmod{2e}$.

Step 2: the piece is a signed sum of sums over $\Gamma_0(e)$ -orbits. The sawtooth form of Lemma 1 applies to the model object, whose moduli are all $d \geq 1$ with $(d, 2D) = 1$: it used neither λ nor squarefreeness, and the modulus $d = 1$, which the lemma excluded, is now kept, because the family $\mathcal{H}_1$ must contain the point $z_{(1,0)} = i\sqrt{|D|}$ to be $\text{SL}_2(\mathbb{Z})$ -invariant. Its single root $x = 0$ contributes $\psi(t)$, and $\#\{h \leq t\} = [t] = t - \frac{1}{2} + \psi(t)$; so, for real $t \geq 0$,

$$S(t) := \sum_{h \leq t} \sigma_{-1}^*(h^2 - D) - t \mathbb{E} \sigma_{-1}^* + \frac{1}{2} = \sum_{(d, 2D)=1} \frac{1}{d} \sum_{x \in R_d} \psi\left(\frac{t-x}{d}\right), \quad \mathbb{E} \sigma_{-1}^* = \sum_{(d, 2D)=1} \frac{\rho(d)}{d^2},$$

the series summed with x and $-x$ paired. Its Riesz means $\mathcal{G}^{(m)}(Y) = \frac{1}{m!} \int_0^Y (Y-t)^m S(t) dt$ are the objects of Theorem 25; at integer Y , $\mathcal{G}^{(0)}(Y) = \int_0^Y S(t) dt$ is exactly the model object (25), linear term included. Its terms are indexed by the pairs (d, x) with $(d, 2D) = 1$, and by Möbius inversion

$$S(t) = \sum_{e \mid \text{rad}(2D)} \mu(e) S_e(t), \quad S_e(t) = \sum_{e \mid d} \frac{1}{d} \sum_{x \in R_d} \psi\left(\frac{t-x}{d}\right),$$

each S_e again summed in pairs and convergent by the tail estimate of Lemma 1: the only unpaired root is $x = 0$ at $d = 1$ (the moduli are odd and coprime to D , so $0 \notin R_d$ and $d/2 \notin \mathbb{Z}$ for $d > 1$). ⁶ Substituting (16), each S_e is a sum over the sub-family $\mathcal{H}_e$ taken modulo translation ($x \bmod d$ is $\Re z \bmod 1$):

$$S_e(t) = \frac{1}{\sqrt{|D|}} \sum_{z \in \mathcal{H}_e / \mathbb{Z}} \Phi_t(z), \quad \Phi_t(z) = \Im z \cdot \psi\left(\frac{t \Im z}{\sqrt{|D|}} + \Re z\right), \quad (17)$$

⁵Checked numerically: for fifteen pairs (D, e) with $e \mid \text{rad}(2D)$ the number of $\Gamma_0(e)$ -orbits met by the forms with $d \leq 60, 120$ and 240 is the same, e.g. 3 for $(-4, 2)$, 2 for $(-3, 3)$, 6 for $(-20, 5)$, 10 for $(-20, 10)$; and 3000 random elements of $\Gamma_0(e)$ preserve $e \mid a$ in every case. $\mathcal{H}_e$ is *not* Γ -stable: for $D = -4$, $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ carries $[5, 4, 1]$ to $[2, 2, 1]$.

⁶The regrouping was checked in exact rational arithmetic for $D = -4$, $t = 37.5$ and 1000.5 , $d \leq 6000$.

since $\frac{1}{d}\psi\left(\frac{t-x}{d}\right) = \frac{\Im z}{\sqrt{|D|}}\psi\left(\frac{t\Im z}{\sqrt{|D|}} + \Re z\right)$ at $z = z_{(d,x)}$. By Lemma 11 each $\mathcal{H}_e$ is a finite union of $\Gamma_0(e)$ -orbits. This is the whole content of the change of route: the object is not a sum over moduli carrying Kloosterman sums, but a signed combination of sums of one fixed test function over orbits of the groups $\Gamma_0(e)$, $e \mid \text{rad}(2D)$, in $\mathbb{H}$, with the length t entering only as a dilation parameter of the test function.

Remark 13 (the coprimality condition is not Γ -invariant: the error of the first version). The first version of this section (12 September 2026) wrote $S(t)$ itself, with its condition $(d, 2D) = 1$, as a sum over complete Γ -orbits of Heegner points of discriminant D , and applied the spectral theory of $\Gamma \backslash \mathbb{H}$. That is wrong: the leading coefficient of a form is not a class invariant (its values over an orbit are the numbers the form represents), and the set of forms with $(a, 2D) = 1$ is a proper non-empty subset of every orbit. For $D = -4$, where $h(D) = 1$, the admitted form $[5, 4, 1]$ is carried by $z \mapsto z/(z+1)$ to the excluded form $[2, 2, 1]$; for the eleven discriminants $-3 \geq D \geq -56$ every class contains forms of both kinds among $a \leq 300$. The error was found by an external assessment of the repository (12 September 2026; [research/reviews/](#)), not by us, and is recorded in the repository's errata (item 13). What *is* invariant is the divisibility $e \mid a$, under $\Gamma_0(e)$ (Lemma 11); hence the Möbius inversion of Step 2 and the passage to the groups $\Gamma_0(e)$, which is the form of the argument now given.

Two consequences are visible in the data and were not predicted by the first version: the spectrum of the model object contains, besides the even Maass forms of level one, the even *newforms* of the levels $e \mid \text{rad}(2D)$; and the unrestricted divisor sum $\sum_{d|h^2-D} d^{-1}$, which is the term $e = 1$ alone and a finite union of Γ -orbits, does not. Both were checked on 13 September 2026, before the repair was written (`scripts/piece-divset.ts`, `piece-level2.ts`, $Y \leq 10^7$; Maass parameters from the LMFDB): after the six level-one even lines are removed, the restricted object and its difference from the unrestricted sum carry the eight smallest even newforms of level e at the 90th–100th percentile against 400 random sets of eight frequencies ($D = -4, -8$ with $e = 2$; $D = -3$ with $e = 3$), the unrestricted sum at the 16th–78th; and for $D = -3$ the largest single line of the restricted object is at 5.085, where the first even Maass newform of level 3 has $t = 5.0987$, while at that frequency the unrestricted sum has $R^2 = 0.01$. Not seen: level-2 lines for $D = -3$, although $2 \mid \text{rad}(2D)$; the corresponding periods may be small, and we record the absence. The same non-invariance, at the level $4u^2$ that grows with u , is the obstruction of §7: the sub-family there is $\{2u^2 \mid b\}$ at discriminant $4u^2D$.

Remark 14 (the Weyl series is a Poincaré-type series). Before going on, it is worth recording what (17) says about the Dirichlet series of the Weyl sums themselves, since it is the reason the spectrum can enter at all. Directly from the coordinates (16), for $\Re s > 1$,

$$\sum_{z \in \mathcal{H}_e/\mathbb{Z}} (\Im z)^s e(m \Re z) = |D|^{s/2} \sum_{e \mid d} \frac{1}{d^s} \sum_{x^2 \equiv D(d)} e\left(\frac{-mx}{d}\right) : \quad (18)$$

the Dirichlet series of the Weyl sums of a quadratic congruence is a Poincaré-type series summed over the Heegner points of discriminant $4D$, modulo translation — over all of them for the unrestricted Weyl sums ($e = 1$), and in the signed combination over e for the Weyl sums over moduli coprime to $2D$. The proof below does not use (18); it works with the test function directly, which is what makes it elementary. We record the identity because it is the structural statement behind everything, and because §6.1 tests the same law on the bare sums it describes.

Step 3: unfolding and the period. One point of bookkeeping decides the shape of everything that follows. The arithmetic parametrisation of (15) is by pairs (d, x) with x taken *modulo* d , and $x \mapsto x + d$ is $z \mapsto z - 1$; so the sum in (17) runs over the points of $\mathcal{H}_e$ *modulo translation*,

that is over $\Gamma_\infty \backslash \Gamma_0(e)z_Q$ and not over $\Gamma_0(e)z_Q$, where $\Gamma_\infty = \{z \mapsto z + n\}$ is the stabiliser of the cusp ∞ , the same subgroup for every e . This is not a normalisation: a full orbit contains $z_Q + n$ for every $n \in \mathbb{Z}$, all of the same height, and $\sum_{\gamma \in \Gamma} (\Im \gamma w)^2$ diverges for that reason, whereas $\sum_{\gamma \in \Gamma_\infty \backslash \Gamma} (\Im \gamma w)^2 = E(w, 2)$ converges. Accordingly we set

$$P_e[\Phi_t](w) = \sum_{\gamma \in \Gamma_\infty \backslash \Gamma_0(e)} \Phi_t(\gamma w),$$

a Poincaré series for $\Gamma_0(e)$ in the classical sense, so that

$$S_e(t) = \frac{1}{\sqrt{|D|}} \sum_{Q \in \Gamma_0(e) \backslash \mathcal{H}_e} \frac{P_e[\Phi_t](z_Q)}{|\Gamma_0(e)z_Q|},$$

the sum over the finitely many orbits of Lemma 11, $\Gamma_0(e)z_Q$ the stabiliser (of order 1, 2 or 3). The spectral decomposition of $P_e[\Phi_t]$ on $L^2(\Gamma_0(e) \backslash \mathbb{H})$, together with the classical unfolding $\langle P_e[\Phi_t], u \rangle = \int_0^\infty \int_0^1 \Phi_t(z) \overline{u(z)} \frac{dx dy}{y^2}$ — the x -integral over a single period, which is exactly the integral computed in Step 4; the cusp ∞ of $\Gamma_0(e)$ has width 1 — gives

$$S_e(t) = \frac{1}{\sqrt{|D|}} \left[(\text{constant and Eisenstein terms}) + \sum_j \langle \Phi_t, u_j^{(e)} \rangle \text{Per}_{D,e}(u_j^{(e)}) \right], \quad (19)$$

$$\text{Per}_{D,e}(u) = \sum_{Q \in \Gamma_0(e) \backslash \mathcal{H}_e} \frac{u(z_Q)}{|\Gamma_0(e)z_Q|},$$

the sum over an orthonormal basis $\{u_j^{(e)}\}$ of the Maass cusp forms of $\Gamma_0(e)$, which may be taken to consist of Hecke eigenforms: the newforms of the levels dividing e and the oldforms they generate, the level-one forms among them. For $e = 1$ the geometric factor is the Katok–Sarnak period of §4, summed over the discriminants $4D/g^2$; for $e > 1$ it is the same kind of sum over the Heegner points of level e . By Corollary 12 it vanishes for every odd $u_j^{(e)}$: the parity rule is now visible as the statement that $\mathcal{H}_e$ is symmetric about the imaginary axis while an odd form is antisymmetric.

Step 4: where the frequencies come from. No continuation theorem is needed here. Expand ψ in Fourier series; since ψ is odd,

$$\psi(a+x) + \psi(a-x) = \frac{2}{\pi} \sum_{k \geq 1} \frac{\sin(2\pi k a) \cos(2\pi k x)}{k}, \quad a = \frac{t \Im z}{\sqrt{|D|}},$$

so that $\Phi_t^+(z) = \frac{y}{\pi} \sum_k k^{-1} \sin(2\pi k a) \cos(2\pi k x)$. Pair this against the Fourier expansion at the cusp ∞ of an even Maass cusp form of $\Gamma_0(e)$, $u_j(z) = 2\sqrt{y} \sum_{n \geq 1} a_j(n) K_{it_j}(2\pi n y) \cos(2\pi n x)$. The x -integral is $\int_0^1 \cos(2\pi k x) \cos(2\pi n x) dx = \frac{1}{2} \delta_{n,k}$, so

$$\langle \Phi_t^+, u_j \rangle = \sum_{k \geq 1} \frac{a_j(k)}{\pi k} I_k, \quad I_k = \int_0^\infty y^{-1/2} \sin(\beta_k y) K_{it_j}(\alpha_k y) dy, \quad \alpha_k = 2\pi k, \quad \beta_k = \frac{2\pi k t}{\sqrt{|D|}}. \quad (20)$$

Substituting $y = w/\beta_k$ gives $I_k = \beta_k^{-1/2} \int_0^\infty w^{-1/2} \sin(w) K_{it_j}(\alpha_k w/\beta_k) dw$, in which $\alpha_k/\beta_k = \sqrt{|D|}/t$ is small. Two standard facts finish the computation: the small-argument expansion

$$K_{i\tau}(x) = \frac{1}{2} \left[\Gamma(-i\tau) \left(\frac{x}{2}\right)^{i\tau} + \Gamma(i\tau) \left(\frac{x}{2}\right)^{-i\tau} \right] + O(x^2), \quad (21)$$

and $\int_0^\infty w^{\sigma-1} \sin w dw = \Gamma(\sigma) \sin(\pi\sigma/2)$ for $0 < \Re \sigma < 1$, applied at $\sigma = \frac{1}{2} \pm it_j$. Hence

$$I_k = \beta_k^{-1/2} \left[\Gamma(-it_j) \Gamma\left(\frac{1}{2} + it_j\right) \sin \frac{\pi(\frac{1}{2} + it_j)}{2} \left(\frac{\sqrt{|D|}}{2t}\right)^{it_j} \cdot \frac{1}{2} + \overline{(\cdot)} \right] + O((\sqrt{|D|}/t)^2). \quad (22)$$

Every t -dependence is now explicit: $\beta_k^{-1/2} \propto t^{-1/2}$ and the bracket carries $t^{\mp it_j}$. So $\langle \Phi_t^+, u_j \rangle = t^{-1/2} (A_j(D) t^{-it_j} + \overline{A_j(D)} t^{it_j}) + O(t^{-5/2})$, and by (19) the same holds for $S_e(t)$ with A_j multiplied by $\text{Per}_{D,e}(u_j)$, hence for $S(t)$ after the signed sum over e . Summing over $t < Y$ turns $t^{-1/2 \mp it_j}$ into $Y^{1/2 \mp it_j}$, which is the observed law.

Two features of the data are read off (22) directly. The frequencies are the t_j themselves, not $2t_j$, because (21) produces $x^{\pm it_j}$ and not $x^{\pm 2it_j}$. And the phase carries the factor $(\sqrt{|D|}/2)^{it_j}$, so that changing the discriminant shifts the phase by exactly $\frac{t_j}{2} \log |D|$ plus the sign of $\text{Per}_D(u_j)$: this is precisely the prediction (26) tested in Tables 5 and 6, and it is what those tables confirm to a median of 0.012 radians. ⁷

Remark 15 (what this route does *not* need). The derivation above uses no continuation theorem for any zeta or Poincaré series: only the Fourier expansions of ψ and of u_j , the small-argument behaviour of the Bessel function, and one elementary integral. This matters, because our two earlier attempts to route the argument through such a theorem both failed. The first, through Selberg's Kloosterman zeta function, predicted the frequencies $2t_j$ and was refuted by the data (Remark 7). The second, through the meromorphic continuation of the Poincaré series of [4], does not apply: their $P_m(z, s)$ carries the factor $e(m\gamma z)$, whose imaginary part contributes an exponential decay $e^{-2\pi m \Im \gamma z}$ and puts the series in L^2 , whereas the series arising in (18) carries only $e(m \Re \gamma z)$ and is not the same object. The identity (18) remains true and remains the reason the spectrum enters; what is not available is a citable continuation of the particular series it produces.

The same computation bears on the size of these sums. Formally, (22) makes every spectral term of $S(t)$ of size $t^{-1/2}$, so the Riesz mean is $O(Y^{1/2+\varepsilon})$ as soon as the sum over j converges: square-root cancellation, against the $X^{3/4+\varepsilon}$ of Hooley [6] and the $X^{2/3+\varepsilon}$ of Bykovskii [1] for the sharp sums. That is exactly what §6.1 observes, with $T_1(X)/\sqrt{X}$ below 0.56, 0.53 and 0.86 for $D = -4, -3, -8$ over $X \leq 10^6$. We do not state it as a proposition, because the convergence of the sum over j is the unfinished step below and there is no shortcut round it: an earlier version of this paper deduced it instead from a continuation theorem that, on inspection, applies to a different series (Remark 15).

Theorem 8 is proved, in the precise form of Theorem 25, in §5.2.

Step 5: why the Poincaré series converges. As it stands $|\Phi_t(z)| \leq \frac{1}{2} \Im z$ and $\sum_{\gamma \in \Gamma_\infty \backslash \Gamma_0(e)} \Im(\gamma w)$ diverges, so $P_e[\Phi_t]$ is not absolutely convergent as it stands. The divergence is exactly that of the Eisenstein series at $s = 1$. It is removed by the symmetry of the sub-family $\mathcal{H}_e$, which we may as well build into the seed.

Lemma 16 (the symmetrised seed decays quadratically). *Write $z = x + iy$, $a = a_t(y) = ty/\sqrt{|D|}$, and $\Phi_t^+(z) = \frac{1}{2}(\Phi_t(z) + \Phi_t(-\bar{z})) = \frac{y}{2}(\psi(a+x) + \psi(a-x))$. Then*

(i) *if $\|x\| > a$, where $\|\cdot\|$ is the distance to $\mathbb{Z}$ and $a < \frac{1}{2}$, then $\Phi_t^+(z) = -ay = -ty^2/\sqrt{|D|}$ exactly;*

(ii) *$|\Phi_t^+(z)| \leq y/2$ always, and the exceptional set $\{x : \|x\| \leq a\}$ has measure $2a$ in x modulo 1.*

Moreover Φ_t^+ may replace Φ_t in (17) without changing the value, because $\mathcal{H}_e$ is stable under $z \mapsto -\bar{z}$ (Lemma 11): the pair (d, x) maps to $(d, -x)$.

Proof. ψ is odd and, on any interval free of integers, satisfies $\psi(\theta + a) = \psi(\theta) - a$. If $\|x\| > a$ then neither x nor $-x$ crosses an integer when shifted by a , so $\psi(x+a) + \psi(-x+a) = \psi(x) +$

⁷Both standard facts were checked numerically; (21) to a relative 10^{-6} at $x \leq 0.1$ for $t = 13.78, 17.74$.

$\psi(-x) - 2a = -2a$, giving (i); (ii) is $|\psi| \leq \frac{1}{2}$ and the definition of the exceptional set. The last statement is $z_{(d,-x)} = -\overline{z_{(d,x)}}$ for $D < 0$. $\square$

So the seed is $O(ty^2)$ off a set of x -measure $O(ty)$, on which it is $O(y)$; both contributions to $\sum_{\gamma \in \Gamma_\infty \setminus \Gamma_0(e)} |\Phi_t^+(\gamma w)|$ are then dominated by $t \sum_{\gamma \in \Gamma_\infty \setminus \Gamma} (\Im \gamma w)^2 = tE(w, 2)$, which converges (the cosets of $\Gamma_0(e)$ are among those of Γ). The arithmetic form of this convergence is the tail estimate already proved in Lemma 1. This is the fourth and last appearance of one symmetry: the root set of a quadratic congruence is stable under $x \mapsto -x$, and that single fact makes the linear term of the sawtooth identity vanish (Lemma 1), kills the odd half of Hurwitz's formula (Proposition 4), annihilates the periods of the odd Maass forms (Proposition 9), and now renders the Poincaré series convergent.

5.1 Completing the decomposition

Three things were left open above: the two conventions for the modulus, the sense in which the spectral decomposition of Step 3 is legitimate, and the order of Riesz mean at which the sum over the spectrum converges. We settle all three.

Lemma 17 (the two conventions). *Let d be odd and coprime to D . Reduction modulo d is a bijection*

$$\{b' \bmod 2d : b'^2 \equiv D \pmod{4d}\} \longrightarrow \{x \bmod d : x^2 \equiv D \pmod{d}\},$$

*so that for such d the pairs of (15) are also parametrised by the forms $[d, b', *]$ of discriminant D , with $e(2kb'/(2d)) = e(kx/d)$. For even d , or $(d, D) > 1$, no such bijection exists in general: for $D \equiv 5 \pmod{8}$ there is no form of discriminant D with even leading coefficient at all. This is why the sub-families of Lemma 11 are taken in discriminant $4D$.*

Proof. Since $(d, 2) = 1$, the Chinese remainder theorem splits $b'^2 \equiv D \pmod{4d}$ into $b'^2 \equiv D \pmod{d}$ and $b'^2 \equiv D \pmod{4}$; the latter is solvable because $D \equiv 0$ or $1 \pmod{4}$ and determines b' modulo 2, which together with b' modulo d determines b' modulo $2d$. The phases agree because $b' \equiv x \pmod{d}$. For the last claim, $b'^2 - 4dc = D$ with d even forces $b'^2 \equiv D \pmod{8}$, and 5 is not a square modulo 8. ⁸ $\square$

Lemma 18 (truncation, and why the decomposition is legitimate). *Every point of $\mathcal{H}_e$ satisfies $\Im z = \sqrt{|D|}/d \leq \sqrt{|D|}$. Consequently, replacing Φ_t^+ by its truncation $\Phi_t^b = \Phi_t^+ \cdot \mathbf{1}_{\{\Im z \leq \sqrt{|D|}\}}$ changes neither side of (17). The truncated seed satisfies*

- (i) $|\Phi_t^b(z)| \leq \sqrt{|D|}/2$, and $\Phi_t^b(z) = 0$ for $\Im z > \sqrt{|D|}$;
- (ii) $\Phi_t^b(z) \ll t(\Im z)^2$ as $\Im z \rightarrow 0$, in the sense of Lemma 16;
- (iii) $\int_0^1 \Phi_t^b(x + iy) dx = 0$ for every y .

Hence $P_e[\Phi_t^b]$ is bounded on $\Gamma_0(e) \setminus \mathbb{H}$, lies in L^2 , and its spectral decomposition is valid.

Proof. (i) and (ii) are Lemma 16 and the truncation. (iii) holds because ψ has mean zero over a period, so $\int_0^1 [\psi(a+x) + \psi(a-x)] dx = 0$ for every y ; the truncation does not depend on x and so preserves this. For the last statement one needs a count, not only the measure statement of Lemma 16(ii), since P_e is a discrete sum (a second reading pointed this out, 13 September 2026). Let w lie in a fundamental domain of Γ , so $\Im w \geq \sqrt{3}/2$, and consider the cosets $\gamma = \begin{pmatrix} * & * \\ c & d \end{pmatrix} \in \Gamma_\infty \setminus \Gamma$ with $\Im \gamma w = \Im w / |cw + d|^2 \in [y, 2y]$, $y \leq \sqrt{|D|}$. They have $c \ll (\Im w / y)^{1/2} =: C_y$ and, for each c , $\ll 1 + C_y$ values of d , so there are $\ll C_y^2 \asymp 1/y$ of them, and $\Re \gamma w = a/c + O(1/(cC_y))$ with

⁸The bijection was verified for $D \in \{-11, -8, -7, -4, -3, 5, 8, 12\}$ and all odd $d \leq 200$ coprime to $2D$.

$ad \equiv 1 \pmod{c}$. The exceptional ones, with $\|\Re \gamma w\| \leq a_t(y) = ty/\sqrt{|D|}$, have $\|a/c\| \ll ty/\sqrt{|D|} + y^{1/2}$, and for each $c \leq C_y$ the number of $a \pmod{c}$ with $\|a/c\| \leq \eta$ is $\ll 1 + \eta c$; summing over c gives $\ll C_y + \eta C_y^2 \ll y^{-1/2} + t/\sqrt{|D|}$ exceptional cosets in the block, each contributing $\leq y$ to the sum, while the $\ll 1/y$ non-exceptional ones contribute $\ll ty^2$ each by Lemma 16(i). The block therefore contributes $\ll y^{1/2} + ty$, and the sum over the dyadic blocks $y \leq \sqrt{|D|}$ is $\ll_D 1 + t$. The cosets of $\Gamma_0(e)$ are among those of Γ , so $P_e[\Phi_t^b]$ is bounded (by $\ll_D 1 + t$) on the quotient, hence square-integrable on the finite-volume $\Gamma_0(e) \backslash \mathbb{H}$. $\square$

Remark 19 (the continuous spectrum). Property (iii) also disposes of the Eisenstein contribution. The group $\Gamma_0(e)$ has finitely many cusps $\mathfrak{a}$ (one for $e = 1$, $2^{\omega(e)}$ for squarefree e), with Eisenstein series $E_{\mathfrak{a}}(z, s)$ whose Fourier expansion at the cusp ∞ is $\delta_{\mathfrak{a}\infty} y^s + \varphi_{\mathfrak{a}\infty}(s) y^{1-s}$ plus non-constant modes $\varphi_{\mathfrak{a}\infty}(n, s) \sqrt{y} K_{s-1/2}(2\pi|n|y) e(nx)$ [9, Theorem 3.4]. Unfolding gives $\langle P_e[\Phi_t^b], E_{\mathfrak{a}}(\cdot, \frac{1}{2} + ir) \rangle = \int_0^\infty \int_0^1 \Phi_t^b \overline{E_{\mathfrak{a}}(\cdot, \frac{1}{2} + ir)} d\mu$, and the two constant terms do not depend on x , so by (iii) they contribute nothing, for every cusp. Only the non-constant modes remain, and against those the computation of Step 4 is the same one with it_j replaced by ir and $a_j(n)$ by the coefficients $\varphi_{\mathfrak{a}\infty}(n, \frac{1}{2} + ir)$, which are of polynomial size in $|n|$ and $|r|$. The continuous spectrum therefore contributes $\sqrt{Y} \int c(r) Y^{ir} dr$ to the Riesz mean, which is $o(\sqrt{Y})$ by the Riemann–Lebesgue lemma as soon as $c \in L^1$, the Riesz factor $\asymp |r|^{-m-1}$ supplying the integrability. This is why the discrete lines are what one sees.

Proposition 20 (which Riesz order). *Normalise u_j in $L^2(\Gamma_0(e) \backslash \mathbb{H})$. For a newform of level $e' \mid e$ write $a_j(n) = \rho_j(1) \lambda_j(n)$ with λ_j the Hecke eigenvalues; an oldform $u(e''z)$ has the coefficients of u at the indices divisible by e'' , and the statements below hold for it with the constants of u . Then in (22)*

$$|A_j(D)| \asymp_D t_j^{-1/2+o(1)}, \quad (23)$$

and the Riesz mean of order m multiplies the j -th term by a factor $\asymp t_j^{-m-1}$. Consequently the spectral sum of Theorem 8 converges absolutely as soon as

$$\begin{aligned} m &\geq 1 \quad (\text{using } \|u_j\|_\infty \ll t_j^{5/12+\varepsilon} \text{ on compact sets, Iwaniec–Sarnak}), \\ \text{or} \quad m &\geq 2 \quad (\text{using only } \|u_j\|_\infty \ll t_j^{1/2+\varepsilon}). \end{aligned}$$

In particular the Cesàro mean, $m = 1$, suffices, which is the mean at which the numerics of §6 were computed.

Proof. For (23): the k -sum $\sum_k a_j(k) k^{-3/2} = \rho_j(1) \sum_k \lambda_j(k) k^{-3/2}$ converges absolutely and is $\rho_j(1) t_j^{o(1)}$, since $\lambda_j(n) \ll n^{7/64+\varepsilon}$ at every level [12] and $\frac{3}{2} - \frac{7}{64} > 1$. The Γ -factors of (22) have modulus

$$|\Gamma(-it_j)| |\Gamma(\tfrac{1}{2} + it_j)| \left| \sin \frac{\pi(\frac{1}{2} + it_j)}{2} \right| \asymp \sqrt{\frac{2\pi}{t_j}} e^{-\pi t_j/2} \cdot \sqrt{2\pi} e^{-\pi t_j/2} \cdot \frac{e^{\pi t_j/2}}{2} \asymp \frac{1}{\sqrt{t_j}} e^{-\pi t_j/2},$$

while $|\rho_j(1)|^2 \asymp_e \cosh(\pi t_j) / L(1, \text{sym}^2 u_j)$ with $t_j^{-\varepsilon} \ll L(1, \text{sym}^2 u_j) \ll t_j^\varepsilon$ [5] gives $|\rho_j(1)| \asymp e^{\pi t_j/2} t_j^{o(1)}$, the implied constants depending on the (fixed) level. The two exponentials cancel and (23) follows. For the Riesz factor, $\frac{1}{m!} \int_0^Y (Y-t)^{m-1/2-it_j} dt = \frac{Y^{m+1/2-it_j}}{m!} B(m+1, \frac{1}{2}-it_j)$ and $B(m+1, \sigma) = \Gamma(m+1)\Gamma(\sigma)/\Gamma(m+1+\sigma) \asymp |\sigma|^{-m-1}$. Finally $|\text{Per}_{D,e}(u_j)| \leq N_{D,e} \|u_j\|_\infty$, with $N_{D,e}$ the number of orbits in Lemma 11 and the supremum over the compact set containing the finitely many points, so the j -th amplitude is $\ll t_j^{5/12+\varepsilon} \cdot t_j^{-1/2+o(1)} \cdot t_j^{-m-1}$, and by the Weyl law for $\Gamma_0(e)$, $\#\{j : t_j \leq T\} \sim \text{vol}(\Gamma_0(e) \backslash \mathbb{H}) T^2/(4\pi)$, the sum over j is dominated by $\int_0^\infty t^{5/12-1/2-m-1} t dt = \int_0^\infty t^{-m-1/12} dt$, which converges for $m > \frac{11}{12}$. With the trivial sup-norm bound the exponent is $\int_0^\infty t^{-m} dt$ instead. $\square$

The expansion (21), uniformly in the order. One point remains: (21) is an asymptotic as $x \rightarrow 0$ for fixed order, whereas we apply it at $x = \sqrt{|D|}w/t$ with t_j ranging over the whole spectrum. What is needed is a version uniform in t_j . It is elementary, and follows from the convergent series for K_ν rather than from any uniform-asymptotic machinery.

Lemma 21 (uniform small-argument expansion). *For $\tau \geq 1$ and $x > 0$,*

$$K_{i\tau}(x) = \frac{1}{2} \left[\Gamma(-i\tau) \left(\frac{x}{2}\right)^{i\tau} + \Gamma(i\tau) \left(\frac{x}{2}\right)^{-i\tau} \right] + E(x, \tau), \quad |E(x, \tau)| \leq |\Gamma(i\tau)| (e^{x^2/(4\tau)} - 1).$$

In particular $|E(x, \tau)| \leq |\Gamma(i\tau)| x^2/(2\tau)$ whenever $x^2 \leq 2\tau$.

Proof. From $K_\nu = \frac{\pi}{2\sin\nu\pi}(I_{-\nu} - I_\nu)$, the series $I_\nu(x) = \sum_{n \geq 0} (x/2)^{2n+\nu}/(n! \Gamma(n+\nu+1))$ and $\Gamma(z)\Gamma(1-z) = \pi/\sin\pi z$,

$$K_{i\tau}(x) = \frac{1}{2} \Gamma(i\tau) \Gamma(1-i\tau) \sum_{n \geq 0} \frac{(x/2)^{2n}}{n!} \left[\frac{(x/2)^{-i\tau}}{\Gamma(n+1-i\tau)} - \frac{(x/2)^{i\tau}}{\Gamma(n+1+i\tau)} \right],$$

convergent for every x . The term $n = 0$ is the stated main term, using $\Gamma(1-i\tau) = -i\tau \Gamma(-i\tau)$. For $n \geq 1$,

$$\left| \frac{\Gamma(1 \mp i\tau)}{\Gamma(n+1 \mp i\tau)} \right| = \frac{1}{|(1 \mp i\tau)(2 \mp i\tau) \cdots (n \mp i\tau)|} \leq \tau^{-n},$$

since each factor has modulus at least τ . As $|\Gamma(i\tau)| = |\Gamma(-i\tau)|$, the tail is at most $|\Gamma(i\tau)| \sum_{n \geq 1} (x^2/4\tau)^n/n! = |\Gamma(i\tau)| (e^{x^2/(4\tau)} - 1)$. $\square$

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Lemma 21 is, however, *not* enough for (22) uniformly in j . In I_k the argument is $x = cw$ with $c = \alpha_k/\beta_k = \sqrt{|D|}/t$, the same for every k ; splitting the w -integral at $W = \sqrt{t_j}/c$ and applying the lemma below W gives an error

$$\ll |\Gamma(it_j)| \frac{c^2}{t_j} \int_0^W w^{3/2} dw \asymp |\Gamma(it_j)| c^{-1/2} t_j^{1/4},$$

against a main term of size $\asymp |\Gamma(it_j)| c^{-1/2}$: a *loss* of $t_j^{1/4}$, and the range $w > W$ is not covered at all. (The first version of this paragraph claimed a saving of $t_j^{-3/4}$; the arithmetic was wrong, as a second reading found on 13 September 2026.) The exact Mellin–Barnes evaluation of Lemma 22 replaces this paragraph; what it does and does not give uniformly in j is discussed in Remark 26. Note that the stationary point of the combined phase, where $t_j d(\log x)/dx$ meets $1/c$, sits at $x = \sqrt{|D|}$, independent of t_j .

5.2 The remaining estimates, and the proof

Two estimates finish the argument: an exact treatment of the integral I_k , and a bound for the tail removed by the truncation. Both are elementary. Throughout, the truncation height of Lemma 18 is taken to be $T = t$ rather than $\sqrt{|D|}$; this is permitted, since any height above $\sqrt{|D|}$ leaves the orbit sum unchanged, and it costs only $\|P_e[\Phi_t^b]\|_2 \ll \sqrt{t}$.

Lemma 22 (the integral, exactly). *Let $\alpha, \beta > 0$, $\tau \geq 1$, and*

$$I(\alpha, \beta, \tau) = \int_0^\infty y^{-1/2} \sin(\beta y) K_{i\tau}(\alpha y) dy.$$

⁹The bound was checked numerically at 30 pairs with $\tau \in \{2, 5, 13.78, 40, 100, 300\}$ and $x \in \{0.5, 1, 2, 4, 8\}$; it holds in every case, with ratio between 0.0002 and 0.995.

Then for any $\varepsilon \in (0, \frac{1}{2})$,

$$I = \beta^{-1/2} \Re \left[\Gamma(-i\tau) \Gamma\left(\frac{1}{2} + i\tau\right) \sin \frac{\pi(\frac{1}{2} + i\tau)}{2} \left(\frac{\alpha}{2\beta}\right)^{i\tau} \right] + O_\varepsilon \left(\beta^{-1/2} e^{-\pi\tau/2} \tau^A \left(\frac{\alpha}{\beta}\right)^{2-\varepsilon} \right)$$

for an absolute constant A , uniformly in α, β, τ .

Proof. The Mellin transforms $\int_0^\infty y^{s-1} \sin(\beta y) dy = \Gamma(s) \sin(\pi s/2) \beta^{-s}$, valid for $0 < \Re s < 1$, and $\int_0^\infty y^{s-1} K_{i\tau}(\alpha y) dy = 2^{s-2} \alpha^{-s} \Gamma(\frac{s+i\tau}{2}) \Gamma(\frac{s-i\tau}{2})$, valid for $\Re s > 0$, give by Parseval's formula for the Mellin transform, for any $c \in (0, \frac{1}{2})$,

$$I = \frac{1}{2\pi i} \int_{(c)} \Gamma(s) \sin \frac{\pi s}{2} \beta^{-s} 2^{-\frac{3}{2}-s} \alpha^{s-\frac{1}{2}} \Gamma\left(\frac{\frac{1}{2}-s+i\tau}{2}\right) \Gamma\left(\frac{\frac{1}{2}-s-i\tau}{2}\right) ds. \quad (24)$$

In $\Re s > 0$ the integrand has poles only where the last two Γ -factors do, namely at $s = \frac{1}{2} \pm i\tau + 2n$, $n \geq 0$; $\Gamma(s)$ contributes none there and $\sin(\pi s/2)$ contributes none at all. Move the contour to $\Re s = \frac{5}{2} - \varepsilon$, crossing exactly the two poles with $n = 0$. Their residues are the displayed main term, and on the new line $|\beta^{-s} \alpha^{s-1/2}| = \beta^{-1/2} (\alpha/\beta)^{2-\varepsilon}$, which is the stated saving. That the shifted integral converges, and that its size is $e^{-\pi\tau/2} \tau^A$ times the above, follows from Stirling's formula: on $\Re s = \frac{5}{2} - \varepsilon$ the two Γ -factors have argument of real part $-1 + \frac{\varepsilon}{2}$ and imaginary parts $\frac{1}{2}(\pm\tau - \Im s)$, so their product decays like $e^{-\frac{\pi}{4}(|\tau - \Im s| + |\tau + \Im s|)} \leq e^{-\pi\tau/2}$ times a power of $1 + |s| + \tau$, while $\Gamma(s) \sin(\pi s/2)$ grows at most like a power times $e^{-\pi|\Im s|/2} \cdot e^{\pi|\Im s|/2}$; the resulting integral in $\Im s$ converges absolutely. ¹⁰ $\square$

Lemma 23 (the truncated tail). *For $\alpha, \beta > 0$, $\tau \geq 1$ and $\alpha T \geq 1$,*

$$\left| \int_T^\infty y^{-1/2} \sin(\beta y) K_{i\tau}(\alpha y) dy \right| \ll \alpha^{-1/2} (\alpha T)^{-1/2} e^{-\alpha T},$$

uniformly in β and τ .

Proof. From $K_\nu(x) = \int_0^\infty e^{-x \cosh u} \cosh(\nu u) du$ one gets, for $\nu = i\tau$, $|K_{i\tau}(x)| \leq \int_0^\infty e^{-x \cosh u} du = K_0(x)$. ¹¹ Hence the integral is at most $\int_T^\infty y^{-1/2} K_0(\alpha y) dy$, and substituting $x = \alpha y$ and using $K_0(x) \ll x^{-1/2} e^{-x}$ for $x \geq 1$ gives the claim. $\square$

The bound is uniform in τ but wasteful in it: $|K_{i\tau}(x)| \leq K_0(x)$ discards the factor $e^{-\pi\tau/2}$ that $K_{i\tau}$ carries for $x < \tau$, and that factor is exactly what the normalisation $|\rho_j(1)| \asymp e^{\pi t_j/2}$ of the Fourier coefficients gives back. So the tail of the j -th coefficient is bounded by $e^{\pi t_j/2} e^{-2\pi k t}$, which does not sum over j . Worse, for $t_j \gg kt^2/\sqrt{|D|}$ the phase $\beta y - t_j \log y$ of the integrand is stationary at $y^* = t_j \sqrt{|D|}/(2\pi k t) > t$, inside the discarded range, and there the tail is of the same order as the main term of Lemma 22. The truncation at height t leaves the orbit sum unchanged, but the estimate of the coefficients with $t_j \gg t^2$ is not the one the proof below uses. This was found by a second reading on 13 September 2026; see Remark 26.

Proposition 24 (the exact coefficient of the Riesz-smoothed seed; added 13 September 2026). *For $m \geq 1$ and $Y > 0$ let $\Psi_Y = \frac{1}{m!} \int_0^Y (Y-t)^m \Phi_t^+ dt$, so that $\frac{1}{m!} \int_0^Y (Y-t)^m S_e(t) dt = \frac{1}{\sqrt{|D|}} \sum_{z \in \mathcal{H}_e/\mathbb{Z}} \Psi_Y(z)$. Then $\Psi_Y(z) = \frac{y}{\pi} \sum_{k \geq 1} k^{-1} \cos(2\pi k x) J_m(c_k y; Y)$ with $c_k = 2\pi k/\sqrt{|D|}$ and $J_m(\omega; Y) = \frac{1}{m!} \int_0^Y (Y-t)^m \sin(\omega t) dt$; $P_e[\Psi_Y]$ is bounded on $\Gamma_0(e) \backslash \mathbb{H}$ (by*

¹⁰The main term was checked against numerical quadrature: for $\tau = 13.7798$ and $\alpha/\beta = 0.1, 0.04, 0.02, 0.01$ the relative error is 0.0715, 0.0099, 0.0018, 0.00035, i.e. $\asymp (\alpha/\beta)^2$ with a bounded constant, confirming both the main term and the exponent.

¹¹Checked at 25 pairs, $\tau \leq 120$, $x \leq 100$: the inequality holds in every case.

$\ll_D Y^{m+2}$) with vanishing x -mean; and for an even Maass cusp form u_j of $\Gamma_0(e)$, written $u_j(z) = 2\sqrt{y} \sum_{n \geq 1} a_j(n) K_{it_j}(2\pi ny) \cos(2\pi nx)$,

$$\begin{aligned} \langle P_e[\Psi_Y], u_j \rangle &= \frac{\sum_{k \geq 1} a_j(k) k^{-3/2}}{\pi \sqrt{2\pi}} \cdot \frac{1}{2\pi i} \int_{(0)} \mathcal{K}_{m,t_j}(s; Y) ds, \\ \mathcal{K}_{m,\tau}(s; Y) &= \frac{\pi |D|^{s/2} Y^{m+1-s} 2^{-s-3/2}}{2 \cos(\pi s/2) \Gamma(m+2-s)} \Gamma\left(\frac{\frac{1}{2}-s+i\tau}{2}\right) \Gamma\left(\frac{\frac{1}{2}-s-i\tau}{2}\right). \end{aligned}$$

Shifting the contour to $\Re s = \frac{3}{4}$ crosses only the poles at $s = \frac{1}{2} \pm it_j$, whose residues are the terms $Y^{m+1/2 \mp it_j}$ of Theorem 25; the integral on $\Re s = \frac{3}{4}$ is $\ll Y^{m+1/4} e^{-\pi t_j/2} t_j^{-5/4} |\sum_k a_j(k) k^{-3/2}| \ll Y^{m+1/4} t_j^{-5/4+\varepsilon}$, uniformly in j .

Proof. The Fourier series is the Riesz mean of the one in Step 4; the bound and the vanishing x -mean follow from Lemma 16 applied for each $t \leq Y$ together with the coset count in the proof of Lemma 18, with Y in place of t ; hence $P_e[\Psi_Y] \in L^2$ and the coefficient is computed by unfolding as in Step 3. For each k use Mellin–Parseval in y with $\int_0^\infty J_m(\omega; Y) \omega^{s-1} d\omega = \pi Y^{m+1-s} / (2 \cos(\pi s/2) \Gamma(m+2-s))$ for $-1 < \Re s < 1$ (from $\int_0^\infty \sin(\omega t) \omega^{s-1} d\omega = t^{-s} \Gamma(s) \sin(\pi s/2)$ and the Beta integral; for $m = 0$ this is the classical $-Y^{1-s} \Gamma(s-1) \sin(\pi s/2)$) and $\int_0^\infty y^{w-1} y^{-1/2} K_{it}(\alpha y) dy = 2^{w-5/2} \alpha^{1/2-w} \Gamma(\frac{w-1/2+it}{2}) \Gamma(\frac{w-1/2-it}{2})$ for $\Re w > \frac{1}{2}$, on a line $\Re s = \sigma \in (-1, \frac{1}{2})$. With $\alpha_k = 2\pi k$ the product $c_k^{-s} \alpha_k^{s-1/2} = |D|^{s/2} (2\pi k)^{-1/2}$ depends on s only through $|D|^{s/2}$, so the k -sum factors out as $\sum_k a_j(k) k^{-3/2}$, absolutely convergent by Kim–Sarnak, which also justifies the interchanges. On the vertical lines the integrand decays like $|\Im s|^{-m-3/4}$, the exponentials of the four Γ -type factors cancelling by Stirling, so the shift to $\Re s = \frac{3}{4}$ is legitimate, and between the two lines the only poles are $s = \frac{1}{2} \pm it_j$: $\cos(\pi s/2)$ first vanishes at $s = 1$, and the next poles of the Γ -factors are at $\frac{5}{2} \pm it_j$. On $\Re s = \frac{3}{4}$, with $\Im s = t$, $|\Gamma(-\frac{1}{8} + \frac{i}{2}(t_j \mp t))| \ll (1 + |t_j \mp t|)^{-5/8} e^{-\pi|t_j \mp t|/4}$, $|\cos(\pi s/2)|^{-1} \ll e^{-\pi|t|/2}$ and $|\Gamma(m + \frac{5}{4} - it)|^{-1} \ll (1 + |t|)^{-m-3/4} e^{\pi|t|/2}$; the product is $\ll e^{-\pi t_j/2} (1 + |t|)^{-m-3/4}$ times $(1 + |t_j - t|)^{-5/8} (1 + t_j + |t|)^{-5/8}$, whose integral over t is $\ll t_j^{-5/4}$, the range $|t| \leq t_j/2$ dominating. ¹² $\square$

Theorem 25 (the spectral formula for the model object; the remainder bound is not established, Remark 26). *Let $D < 0$ be a discriminant, $u = 1$, and let $\mathcal{G}^{(m)}(Y) = \frac{1}{m!} \int_0^Y (Y-t)^m S(t) dt$ be the Riesz mean of order m of the sawtooth series S of Step 2, whose order-0 mean is the model object (25) at integer Y . For each $e \mid \text{rad}(2D)$ let $\{u_j^{(e)}\}$ be an orthonormal basis of the even Maass cusp forms of $\Gamma_0(e)$, with Laplace eigenvalues $\frac{1}{4} + t_{j,e}^2$. For $m \geq 2$ there are coefficients $\alpha_{j,e}(D)$, proportional to the periods $\text{Per}_{D,e}(u_j^{(e)})$ of (19), such that*

$$\mathcal{G}^{(m)}(Y) = Y^m \sqrt{Y} \sum_{e \mid \text{rad}(2D)} \mu(e) \sum_j (\alpha_{j,e}(D) Y^{it_{j,e}} + \overline{\alpha_{j,e}(D)} Y^{-it_{j,e}}) + \mathcal{E}(Y) + O(Y^{m+\frac{1}{2}-\delta})$$

for some $\delta > 0$, each inner sum converging absolutely, where $\mathcal{E}$ is the contribution of the continuous spectra and satisfies $\mathcal{E}(Y) = o(Y^{m+1/2})$. The spectral parameters that occur are those of the even Maass cusp forms of level one and of the even newforms of the levels $e \mid \text{rad}(2D)$, and no others.

¹²Validated numerically before this was written: the closed form of J_m against quadrature; the Mellin–Parseval representation against a direct quadrature with 1400 subintervals, to 8 digits, for $(t_j, Y) = (13.7798, 50)$ and $(5, 20)$; the identity “line at $\Re s = 0$ equals line at $\Re s = \frac{3}{4}$ minus the two residues” to 7 digits for $t_j = 13.78, 40$ and $Y = 50, 5000$; and the decay of the remainder in t_j and its growth $\asymp Y^m$ in Y .

Proof. By Step 2, $S(t) = \sum_e \mu(e) S_e(t)$ is a finite signed sum and the Riesz mean is linear, so it suffices to treat one S_e , given by (17) as a sum over the finitely many $\Gamma_0(e)$ -orbits of $\mathcal{H}_e$ (Lemma 11). Apply the Riesz mean of order m to (17) and truncate the seed at height $T = t$ (Lemma 18), which changes nothing. By Lemma 16 the automorphised seed $P_e[\Phi_t^b]$ converges absolutely and lies in $L^2(\Gamma_0(e)\backslash\mathbb{H})$, so it has a spectral decomposition; by Proposition 20 the Riesz factor $\asymp t_j^{-m-1}$ makes the discrete part converge absolutely for $m \geq 2$ even with only the trivial sup-norm bound, so the expansion converges uniformly and may be evaluated at the points z_Q and summed over the orbits, which replaces $u_j^{(e)}$ by $\text{Per}_{D,e}(u_j^{(e)})$ (Step 3); the odd forms drop out by Corollary 12. The constant term vanishes because $\int_0^1 \Phi_t^b dx = 0$ (Lemma 18(iii)), and the same identity confines the Eisenstein contribution of every cusp to the non-constant modes, whence $\mathcal{E}(Y) = o(Y^{m+1/2})$ by Remark 19. The last sentence of the statement holds because the basis may be chosen to consist of newforms of levels $e' \mid e$ and their oldform images, whose spectral parameters are those of the newforms.

For the coefficients, (20) expresses $\langle \Phi_t^b, u_j \rangle$ as $\sum_k (a_j(k)/\pi k)$ times $I(\alpha_k, \beta_k, t_j)$ minus the truncated tail, with $\alpha_k = 2\pi k$ and $\beta_k = 2\pi kt/\sqrt{|D|}$. The interchange of the k -sum with the y -integral is justified by $\int_0^\infty y^{-1/2} K_0(\alpha_k y) dy \ll k^{-1/2}$ together with $\sum_k |a_j(k)| k^{-3/2} < \infty$, which holds because $\lambda_j(n) \ll n^{7/64+\varepsilon}$. Lemma 22 evaluates each $I(\alpha_k, \beta_k, t_j)$ with a relative error $O(\tau^A(\alpha_k/\beta_k)^{2-\varepsilon})$; crucially $\alpha_k/\beta_k = \sqrt{|D|}/t$ does not depend on k , so the relative error is uniform in k and survives the k -sum unchanged. Collecting the main terms gives, for the untruncated integrals, $\sum_k (a_j(k)/\pi k) I(\alpha_k, \beta_k, t_j) = t^{-1/2} (A_j(D) t^{-it_j} + \overline{A_j(D)} t^{it_j}) + O(t^{-5/2+\varepsilon} t_j^A)$ with $|A_j(D)| \asymp_D t_j^{-1/2+o(1)}$ by (23); the factor $(\alpha_k/2\beta_k)^{it_j} = (\sqrt{|D|}/2t)^{it_j}$ supplies the dependence on the discriminant. Summing $t^{-1/2 \mp it_j}$ against the Riesz weight turns it into $Y^{m+1/2 \mp it_j}$ times $B(m+1, \frac{1}{2} \mp it_j) \asymp t_j^{-m-1}$, which is the main term of the statement, absolutely convergent over j for $m \geq 2$ by Proposition 20.

What this argument does *not* establish is the bound on the remainder. Three things are missing, found by a second reading on 13 September 2026 and listed in Remark 26: the error $O(t^{-5/2+\varepsilon} t_j^A)$ is not oscillatory in t , so the Riesz mean gives it $Y^{m-3/2} t_j^A$ with no decay in j , and $\sum_j t_j^A |\text{Per}(u_j)|$ diverges by the Weyl law for every $A \geq 0$; the truncated tail is not $O(e^{-2\pi t})$ after the sum over j (see after Lemma 23); and at fixed t the spectral sum $\sum_j \langle \Phi_t^b, u_j \rangle u_j(z_Q)$ is not absolutely convergent, so the Riesz mean must be applied to the seed before the expansion, which the truncation at the t -dependent height $T = t$ does not respect as written. The first version of this proof claimed all three (“the error terms sum by Proposition 20 with room to spare”). $\square$

Remark 26 (what is proved, what is not, and why; rewritten 13 September 2026, evening). Proved: the exact identity $S(t) = \sum_e \mu(e) S_e(t)$ with each S_e a sum over the finitely many $\Gamma_0(e)$ -orbits of $\mathcal{H}_e$ (Steps 1–3, Lemma 11); the vanishing of the constant terms; $P_e[\Phi_t^b], P_e[\Psi_Y] \in L^2$; the exact coefficient formula of Proposition 24, hence the main terms $Y^{m+1/2 \pm it_j}$ with the residue coefficients, absolutely summable over j for $m \geq 2$.

Not proved: that the remainder is $o(Y^{m+1/2})$. The per-form remainder of Proposition 24 is $\ll Y^{m+1/4} t_j^{-5/4+\varepsilon}$; with $|\text{Per}_{D,e}(u_j)| \ll t_j^{1/2}$ and the Weyl law the sum over j of these bounds diverges (it is $\sum_j t_j^{-3/4}$ against the density $t_j dt_j$). The exponent $-5/4$ is not improved by the Riesz order, because it comes from the range $|\Im s| \ll t_j$ on the shifted line, where the only decay in t_j is that of the two Γ -factors; and it is not improved by shifting further right, because $1/\cos(\pi s/2)$ has a pole at $s = 1$. That pole is the obstruction, and it has a meaning. Its residue is a term $Y^m r_j$ with $r_j \asymp t_j^{-3/2+o(1)} \text{Per}_{D,e}(u_j)$, whose spectral sum converges at best conditionally; on the geometric side it is the coefficient of Y^m in $\Psi_Y(z)$ for $y \gg \sqrt{|D|}/(kY)$, namely $\frac{\sqrt{|D|}}{2} B_2(\{x\})$ with B_2 the second Bernoulli polynomial — a function of x alone, the memory of the sharp start

of the sum at $h = 1$, which no smoothing at the upper end removes. Summed over the points with $d \lesssim Y$ it is $\frac{Y^m}{2} \sum_{d \lesssim Y} \sum_{x \in R_d} B_2(x/d) = \frac{Y^m}{2\pi^2} \sum_k k^{-2} \sum_{d \lesssim Y} \rho_k(d)$: a Riesz-smoothed form of Hooley's Weyl sums $T_k(Y)$ of §6.1, of size $Y^{1/2}$ conjecturally and $Y^{2/3+\varepsilon}$ provably (Bykovskii), oscillating at the same t_j (§6.1). So the coefficient of $Y^{m+1/2 \pm it_j}$ in $\mathcal{G}^{(m)}(Y)$ is the residue of Proposition 24 *plus* the contribution of these Hooley sums, and a bound for the remainder requires the modulus-side spectral theory (Kuznetsov, Bykovskii) for the latter. This is not a technicality; the first version of this remark called it one, and the analysis above, carried out the same day, shows why it is not. It also offers an explanation of the one prediction of this paper that the data did not confirm, the amplitude law of §6: the amplitudes we predicted were the residues alone.

What *is* provable by exactly this computation is the smooth-window statement. For $w \in C_c^\infty((0, \infty))$, Poisson summation turns $\sum_h w(h/Y) (\sigma_{-1}^*(h^2 - D) - \mathbb{E}\sigma_{-1}^*)$ into a sum over $\mathcal{H}_e/\mathbb{Z}$ of the seed $\frac{2Yy^2}{|D|} \sum_k \Re \hat{w}(kYy/\sqrt{|D|}) \cos(2\pi kx)$, Schwartz in Yy ; its Mellin transform in y is entire to the right and rapidly decaying, so the contour can be moved to $\Re s = \frac{5}{2} - \varepsilon$, where the Γ -factors give t_j^{-3} and the spectral sum converges absolutely with the trivial sup-norm bound. The result has the shape $Y^{-1/2} \sum_e \mu(e) \sum_j (\beta_{j,e} Y^{it_j} + \overline{\beta_{j,e}} Y^{-it_j}) + O(Y^{-3/2+\varepsilon})$: it concerns the increments of $S(t)$ over $[Y/2, Y]$, not $S(t)$ itself, and so does not see the Hooley part. Its shape was checked before this paragraph was written (`scripts/piece-divset.ts`, `SMOOTH=1`): for $D = -4$, $Y \leq 2 \cdot 10^6$, $\sqrt{Y}$ times the smooth-window sum is explained by the six level-one even parameters at the 100th percentile against random frequency sets (the odd set at the 13th–22nd), and for the restricted object its largest line is at 8.95, the first even newform of level 2. The proof is written out below, as Theorem 27.

5.3 The smooth-window theorem

Theorem 27 (the spectral formula for smooth windows). *Let $D < 0$ be a discriminant, $w \in C_c^\infty((0, \infty))$ real, $W(\zeta) = \int_0^\infty w(u) u^{\zeta-1} du$ and $W_c(z) = (2\pi)^{-z} \Gamma(z) \cos(\pi z/2) W(1-z)$. For $e \mid \text{rad}(2D)$ let $\{u_j\}$ be an orthonormal basis of the even Maass cusp forms of $\Gamma_0(e)$, $u_j(z) = 2\sqrt{y} \sum_{n \geq 1} a_j(n) K_{it_j}(2\pi ny) \cos(2\pi nx)$, and put $\tilde{L}_j = \sum_{k \geq 1} a_j(k) k^{-3/2}$ and $\text{Per}_{D,e}(u_j) = \sum_{Q \in \Gamma_0(e) \backslash \mathcal{H}_e} u_j(z_Q) / |\Gamma_0(e)_{z_Q}|$ as in (19). Then for $Y \geq 1$ and every $\varepsilon > 0$,*

$$\begin{aligned} \sum_{h \geq 1} w\left(\frac{h}{Y}\right) (\sigma_{-1}^*(h^2 - D) - \mathbb{E}\sigma_{-1}^*) &= \frac{1}{|D|^{1/4} \sqrt{Y}} \sum_{e \mid \text{rad}(2D)} \mu(e) \sum_j \text{Per}_{D,e}(u_j) \tilde{L}_j [A_j(Y) + \overline{A_j(Y)}] \\ &\quad + \mathcal{E}^w(Y) + O_{w,D,\varepsilon}(Y^{-5/2+\varepsilon}), \\ A_j(Y) &= \Gamma(-it_j) W_c\left(\frac{3}{2} + it_j\right) \left(\frac{\pi \sqrt{|D|}}{Y}\right)^{it_j}, \end{aligned}$$

the sum over j converging absolutely, where $\mathcal{E}^w$ is the contribution of the continuous spectra and satisfies $\mathcal{E}^w(Y) \ll_A Y^{-1/2} (\log Y)^{-A}$ for every A .

Proof. Poisson summation. Write $E = \mathbb{E}\sigma_{-1}^* = \sum_{(d,2D)=1} \rho(d)/d^2$ and $\hat{w}(\xi) = \int w(u) e(-\xi u) du$. Since w has compact support in $(0, \infty)$, $\sum_{h \geq 1} w(h/Y) = \sum_{h \in \mathbb{Z}} w(h/Y) = Y \sum_k \hat{w}(kY) = Y \hat{w}(0) + O(Y^{-N})$, and for each of the finitely many h in the support $\sigma_{-1}^*(h^2 - D) = \sum_{(d,2D)=1} d^{-1} \mathbf{1}_{d|h^2-D}$; so, with $\sum_{h \equiv x \pmod{d}} w(h/Y) = (Y/d) \sum_{k \in \mathbb{Z}} \hat{w}(kY/d) e(kx/d)$ and the $k = 0$ terms summing to $YE \hat{w}(0)$,

$$\sum_h w(h/Y) (\sigma_{-1}^*(h^2 - D) - E) = \sum_{(d,2D)=1} \frac{Y}{d^2} \sum_{k \neq 0} \hat{w}\left(\frac{kY}{d}\right) \rho_k(d) + O(Y^{-N}),$$

$\rho_k(d) = \sum_{x \in R_d} e(kx/d)$, real. (The d -sum is finite for the terms with a lattice point in the support and absolutely convergent for the rest, since $\hat{w}$ is Schwartz.) In the coordinates (16), with $\kappa = Yy/\sqrt{|D|}$ so that $Y/d^2 = Yy^2/|D|$, this is $\sum_e \mu(e) \sum_{z \in \mathcal{H}_e/\mathbb{Z}} \Psi_Y^w(z)$ with the seed

$$\Psi_Y^w(z) = \frac{Yy^2}{|D|} \sum_{k \neq 0} \hat{w}(k\kappa) e(-kx) = \frac{y}{\sqrt{|D|}} \sum_{n \in \mathbb{Z}} w\left(\frac{(n-x)\sqrt{|D|}}{Yy}\right) - \frac{Yy^2}{|D|} \hat{w}(0),$$

the second form again by Poisson. (The double sum over (d, k) is not absolutely convergent — it is $\asymp \sum_d \rho(d)/d$ in absolute value — so the regrouping into the sub-families $\mathcal{H}_e$ and into orbits is carried out on the k -summed second form, where each term is bounded and the d -sum converges as explained.) The seed has vanishing x -mean. For $\kappa \geq 1$ it is $\ll_N (Yy^2/|D|)\kappa^{-N}$: the moduli $d \leq Y^{1-\delta}$, where $\kappa \geq Y^\delta$, contribute $O(Y^{-N})$ in total, while the moduli $d \asymp Y$ do contribute. For $\kappa < 1$ it is $-(Yy^2/|D|)\hat{w}(0)$ off the set $\{x : \exists n, (n-x)/\kappa \in \text{supp } w\}$, of measure $\ll \kappa$, and $\ll y$ on it. So for $\Im \gamma w$ in a dyadic block $[y_0, 2y_0]$ with $Yy_0 < \sqrt{|D|}$ the coset count in the proof of Lemma 18, with Y in place of t , bounds the block by $\ll y_0^{1/2} + Yy_0$, and the blocks with $Yy_0 \geq \sqrt{|D|}$ are bounded by the rapid decay; hence $P_e[\Psi_Y^w]$ converges absolutely and uniformly on compacta, is bounded on $\Gamma_0(e) \backslash \mathbb{H}$ (by $\ll_D 1$, uniformly in Y), and lies in L^2 ; its spectral expansion has no constant term. It is also continuous, Ψ_Y^w being continuous.

The coefficients. Unfolding as in Step 3 and pairing the Fourier series of Ψ_Y^w with that of u_j , $\int_0^1 e(-kx) \cos(2\pi nx) dx = \frac{1}{2} \delta_{|k|,n}$ gives

$$\langle P_e[\Psi_Y^w], u_j \rangle = \frac{2Y}{|D|} \sum_{k \geq 1} a_j(k) \int_0^\infty y^{1/2} \Re \hat{w}(\kappa_k y) K_{it_j}(2\pi ky) dy, \quad \kappa_k = \frac{kY}{\sqrt{|D|}}.$$

Since $\Re \hat{w}(v) = \int w(u) \cos(2\pi uv) du$ and $\int_0^\infty v^{z-1} \cos(2\pi uv) dv = (2\pi u)^{-z} \Gamma(z) \cos(\pi z/2)$ for $0 < \Re z < 1$, the Mellin transform of $y \mapsto y^{1/2} \Re \hat{w}(\kappa y)$ is $\kappa^{-s-1/2} W_c(s + \frac{1}{2})$ for $-\frac{1}{2} < \Re s < \frac{1}{2}$; W_c continues to an entire function on $\Re z > 0$, and on vertical lines $|W_c(\sigma + i\tau')| \ll (1 + |\tau'|)^{\sigma-1/2} |W(1 - \sigma - i\tau')| \ll_N (1 + |\tau'|)^{-N}$, because w is smooth. With $\int_0^\infty y^{v-1} K_{it}(\alpha y) dy = 2^{v-2} \alpha^{-v} \Gamma(\frac{v+it}{2}) \Gamma(\frac{v-it}{2})$ ($\Re v > 0$) and Mellin–Parseval on $\Re s = 0$,

$$\begin{aligned} \langle P_e[\Psi_Y^w], u_j \rangle &= \frac{2Y}{|D|} \tilde{L}_j \cdot \frac{1}{2\pi i} \int_{(0)} \mathcal{K}_{t_j}^w(s; Y) ds, \\ \mathcal{K}_\tau^w(s; Y) &= \left(\frac{Y}{\sqrt{|D|}} \right)^{-s-1/2} W_c(s + \tfrac{1}{2}) 2^{-1-s} (2\pi)^{s-1} \Gamma\left(\frac{1-s+i\tau}{2}\right) \Gamma\left(\frac{1-s-i\tau}{2}\right), \end{aligned}$$

the k -dependence $\kappa_k^{-s-1/2} (2\pi k)^{s-1} = k^{-3/2} (Y/\sqrt{|D|})^{-s-1/2} (2\pi)^{s-1}$ having factored out into $\tilde{L}_j$, which converges absolutely by Kim–Sarnak and justifies the interchanges. The integrand is entire in $\Re s > -\frac{1}{2}$ except for the poles of the Γ -factors at $s = 1 \pm it_j + 2n$, and decays faster than any power of $|\Im s|$ on vertical lines (the exponentials of the Γ -factors cancel by Stirling, W_c supplies the decay). Move the contour to $\Re s = 3 - \varepsilon$; the integral on $\Re s = 0$ equals the integral on $\Re s = 3 - \varepsilon$ minus the residues at $s = 1 \pm it_j$, and with $\Gamma(\frac{1-s \pm it_j}{2}) \sim -2/(s-1 \mp it_j)$ minus those residues is

$$\begin{aligned} &\frac{2Y}{|D|} \tilde{L}_j \left(\frac{Y}{\sqrt{|D|}} \right)^{-3/2} \left[2 \cdot 2^{-2-it_j} (2\pi)^{it_j} \left(\frac{Y}{\sqrt{|D|}} \right)^{-it_j} W_c\left(\frac{3}{2} + it_j\right) \Gamma(-it_j) + \text{c.c.} \right] \\ &= \frac{\tilde{L}_j}{|D|^{1/4} \sqrt{Y}} \left[\Gamma(-it_j) W_c\left(\frac{3}{2} + it_j\right) \left(\frac{\pi \sqrt{|D|}}{Y} \right)^{it_j} + \text{c.c.} \right], \end{aligned}$$

and on $\Re s = 3 - \varepsilon$ the integrand is $\ll Y^{-7/2+\varepsilon} e^{-\pi t_j/2} (1+t_j)^{-3+\varepsilon} (1+|\Im s|)^{-N}$ away from $|\Im s| \approx t_j$, and $\ll Y^{-7/2+\varepsilon} e^{-\pi t_j/2} t_j^{-3/2} (1+|\Im s|)^{-N}$ there (one Γ -factor is then $O(1)$), which with

N large is again $\ll e^{-\pi t_j/2} t_j^{-3}$ after integration; so the integral is $\ll Y^{-5/2+\varepsilon} e^{-\pi t_j/2} t_j^{-3+\varepsilon} |\tilde{L}_j|$ after the prefactor. Here $|\tilde{L}_j| \leq |\rho_j(1)| \sum_k |\lambda_j(k)| k^{-3/2} \ll e^{\pi t_j/2} t_j^\varepsilon$ by Hoffstein–Lockhart and Kim–Sarnak for newforms; for the oldform space of a newform f of level $e' \mid e$ we use the orthonormal basis obtained from $\{f(dz) : d \mid e/e'\}$ by Gram–Schmidt, whose Fourier coefficients are finite combinations of $\rho_f(1)\lambda_f(k/d)$ with coefficients bounded uniformly in f , because the Gram matrix of $\{f(dz)\}$ is a fixed-size matrix with entries polynomial in the $\lambda_f(p)$, $p \mid e$, and determinant bounded below by Kim–Sarnak ($|\lambda_f(p)| \leq 2p^{7/64}$); this is the treatment of oldforms in [10], for Maass forms verbatim. Meanwhile $|\Gamma(-it_j)W_c(\frac{3}{2}+it_j)| \ll t_j^{-1/2} e^{-\pi t_j/2} \cdot t_j |W(-\frac{1}{2}-it_j)| \ll_N t_j^{-N} e^{-\pi t_j/2}$.

Summation and the continuous spectrum. With $|\text{Per}_{D,e}(u_j)| \ll_{D,e} \|u_j\|_\infty \ll t_j^{1/2}$ on the compact set containing the finitely many points, the main terms are $\ll t_j^{-N+1}$ and the remainders $\ll Y^{-5/2+\varepsilon} t_j^{-5/2+2\varepsilon}$, both summable against the Weyl law; so the spectral series converges absolutely and uniformly on compacta, and may be evaluated at the points z_Q . For each cusp $\mathfrak{a}$ of $\Gamma_0(e)$ the same computation with the Fourier coefficients $\varphi_{\mathfrak{a}\infty}(n, \frac{1}{2}+ir)$ of $E_{\mathfrak{a}}$, which are $\ll n^\varepsilon e^{\pi r/2} (1+r)^\varepsilon$, gives $\mathcal{E}^w(Y) = Y^{-1/2} \int c(r) Y^{ir} dr + O(Y^{-5/2+\varepsilon})$. Here the two residues at $s = 1 \pm ir$ combine, since $\overline{E_{\mathfrak{a}}(z, \frac{1}{2}+ir)} = E_{\mathfrak{a}}(z, \frac{1}{2}-ir)$, into a function of r that is real-analytic also at $r = 0$, where the poles of $\Gamma(\mp ir)$ cancel between the two terms; c is therefore smooth, and it and all its r -derivatives are integrable, because they inherit the rapid decay of $W_c(\frac{3}{2}+ir)$ and of its derivatives while $E_{\mathfrak{a}}(z_Q, \frac{1}{2}+ir)$ and its r -derivatives grow at most polynomially in $|r|$ on compacta. Hence $\mathcal{E}^w(Y) \ll_A Y^{-1/2} (\log Y)^{-A}$ by integrating by parts A times in r . The odd forms are absent by Corollary 12, and the sum over e is finite. $\square$

The test, done before the theorem was written. The theorem predicts the phase of each line with no free parameter. For the unrestricted divisor sum at $D = -4$ (the case $e = 1$ alone, $\mathcal{H}_1$ the two orbits of discriminant -16 , points $2i$ and i with stabilisers of order 1 and 2) and $w(u) = \exp(-1/(1-(4u-3)^2))$ on $(\frac{1}{2}, 1)$, the line $t_1 = 13.7798$ of $\sqrt{Y} \sum_h w(h/Y)(\sigma_{-1}(h^2+4)-E)$ should be $C \cos(t_1 \log Y - \varphi)$ with $\varphi = t_1 \log(\pi \sqrt{|D|}) + \arg(\Gamma(-it_1)W_c(\frac{3}{2}+it_1)) + \pi \mathbf{1}[\text{Per } \tilde{L}_1 < 0]$, where $\text{Per} \propto u_1(2i) + u_1(i)/2 > 0$ and $\tilde{L}_1 = 1.866 > 0$ from the LMFDB coefficients of u_1 . Predicted $\varphi = 4.066$; observed, by least squares on $Y \leq 2 \cdot 10^6$, $\varphi = 4.032$: a difference of 0.035 radians, where the opposite sign of the period would give 3.1 (`scripts/smooth-phase-test.py`). The amplitude was then tested as well: the L^2 -norm of u_1 over the fundamental domain, computed from the same twenty coefficients (converged to twenty digits with twelve), gives the normalisation, and the predicted amplitude $2|D|^{-1/4} |\text{Per}(u_1) \tilde{L}_1| |\Gamma(-it_1)W_c(\frac{3}{2}+it_1)| = 0.0600$ against the observed 0.0569 (`scripts/smooth-amplitude-test.py`). Extending the data to $Y \leq 10^7$ and fitting the first six even lines jointly (`scripts/smooth-two-lines.py`) sharpens this to a ratio 0.99 and a phase difference of 0.010 radians for t_1 over the whole range (0.998 and 0.0001 over its upper half), and gives for the second even form, $t_2 = 17.7386$, with its own coefficients from the LMFDB, a predicted amplitude 0.0085 and phase 5.860 against 0.0090 and 5.753 observed over the upper half of the range (the line is seven times weaker than the first and is not resolved over the lower half). Two independent forms, every constant.

The level structure was tested on the restricted object (25) itself, which is the difference of the terms $e = 1$ and $e = 2$ (`scripts/smooth-level2-test.py`). The sub-family $\mathcal{H}_2$ (forms of discriminant -16 with even leading coefficient) consists of three $\Gamma_0(2)$ -orbits, represented by $[2, 0, 2]$, $[4, 0, 1]$ and $[4, 4, 2]$ with stabilisers of orders 1, 1, 2. (i) The line of the first even newform of level 2, $t = 8.9229$ (LMFDB 2.0.1.4.1, Fricke eigenvalue $+1$), occurs only in the term $e = 2$, with the sign $\mu(2) = -1$; its period over the three points, its coefficients and its norm on $\Gamma_0(2) \backslash \mathbb{H}$ (computed over the fundamental domain $F \cup S(F) \cup ST(F)$, the Fricke involution giving the values near the cusp 0) predict amplitude 0.0398 and phase 5.971; observed, 0.0406 and 6.020.

In the unrestricted object the theorem predicts the line absent, and the fit gives 0.0009. (ii) The line t_1 receives, in the restricted object, the $e = 1$ residue *minus* the contribution of the two-dimensional oldform space spanned by $u_1(z)$ and $u_1(2z)$, orthonormalised on $\Gamma_0(2)\backslash\mathbb{H}$ (Gram matrix computed numerically: $\langle u_1(2z), u_1 \rangle / \|u_1\|^2 = 0.730$); predicted amplitude 0.0268 against 0.0600 for the unrestricted object, phase unchanged; observed 0.0271 and 0.0597, phases 4.033 and 4.066. So the Möbius resolution of the coprimality condition, the level-two newforms and the oldform bookkeeping of the theorem are each seen in the data with their predicted sizes and signs. This is the first amplitude prediction of this paper that the data confirm; the amplitude law for the Riesz object in §6 remains unconfirmed, for the reason given in Remark 26.

The amplitude law across discriminants. The amplitude of the line t_1 predicted by Theorem 27 for the unrestricted object is $C(D) = 2|D|^{-1/4}|\text{Per}_D(u_1)\tilde{L}_1| |\Gamma(-it_1)W_c(\frac{3}{2} + it_1)|/\|u_1\|^2$ with $\text{Per}_D(u_1) = \sum_Q u_1(z_Q)/|\Gamma_Q|$ over all classes of forms of discriminant $4D$, primitive or not; it has no free parameter and depends on D through the Heegner values alone. Table 1 compares it with the fitted amplitude on smooth-window data to $Y = 10^7$ for fifteen discriminants (`research/explore/smooth-Dsweep.py`). For the ten discriminants with $|D| \leq 23$ the ratio lies within 5% of 1 and the phase within 0.1 rad, with class numbers up to 6; beyond the turning point $|D| \approx 19$ of the threshold paragraph the predicted amplitude falls to the noise floor of a single line on these grids (about 0.003–0.005) and the ratios scatter accordingly, at $D = -43$ the prediction (0.0006) being below it, as that paragraph anticipated. Adding four further level-one lines to the fit changes the ratios by at most 5%.

Table 1: The line t_1 across discriminants, unrestricted object, smooth window, $Y \leq 10^7$: predicted amplitude $C(D)$, observed amplitude, their ratio, and the phase difference (rad). h is the number of classes of forms of discriminant $4D$ (all forms).

D	h	C_{pred}	C_{obs}	ratio	$\Delta\varphi$
−3	2	0.0406	0.0414	1.02	−0.04
−4	2	0.0600	0.0587	0.98	−0.01
−7	2	0.0307	0.0311	1.01	+0.03
−8	3	0.0321	0.0338	1.05	+0.10
−11	4	0.0410	0.0411	1.00	+0.03
−15	4	0.0609	0.0597	0.98	+0.05
−19	4	0.0526	0.0529	1.01	−0.01
−20	6	0.0514	0.0527	1.03	−0.02
−23	6	0.0331	0.0322	0.97	+0.08
−24	6	0.0161	0.0204	1.27	+0.13
−27	6	0.0100	0.0118	1.18	+0.40
−31	6	0.0075	0.0108	1.45	−0.35
−35	8	0.0209	0.0255	1.22	−0.09
−39	8	0.0069	0.0047	0.68	−0.26
−43	4	0.0006	0.0024	—	—

Remark 28 (what the smooth theorem says and does not say). Since the seed is $O(Y^{-N})$ on the moduli $d \leq Y^{1-\delta}$ (not on all of $d \leq Y$: the moduli $d \asymp Y$ contribute, a point a second reading made precise), Theorem 27 is a statement about the moduli $d \geq Y^{1-\delta}$, essentially the window and the far moduli of part III, for $u = 1$ and the model weights: they oscillate at the Maass parameters, at the scale $Y^{-1/2}$ appropriate to a sum over $h \in [Y/2, Y]$, with the explicit amplitudes. It is the increment of the sharp sum $S(t)$ over a window, and increments do not see the Hooley part of Remark 26, which is why the sharp start of $S(t)$ costs the remainder bound in Theorem 25 and costs nothing here. For the pieces themselves (weight λ , squarefree moduli)

the same theorem holds at each level $m\ell^2$ after the Möbius expansions of §6; the sum over m and ℓ needs the levels to be uniform, a milder form of the problem of §7.

Remark 29 (what a referee should check). The proof uses, as standard inputs: Gauss’s parametrisation of the root pairs by binary quadratic forms of discriminant $4D$ (§5, Step 1), the spectral decomposition of $L^2(\Gamma_0(e)\backslash\mathbb{H})$ for the finitely many levels $e \mid \text{rad}(2D)$ with its Eisenstein series at each cusp, the Weyl law and the sup-norm bound for Maass forms at fixed level, the Kim–Sarnak bound $\lambda_j(n) \ll n^{7/64+\varepsilon}$ and the Hoffstein–Lockhart normalisation of $\rho_j(1)$ for newforms of fixed level, and Parseval’s formula for the Mellin transform. The step that was wrong in the first version, the passage from the restricted sum to orbits (Remark 13), was found by an external reader; its replacement is Lemma 11 together with Step 2. A second reading then found the error bookkeeping of the remainder insufficient (Remark 26); that is now the open part of the proof. The two places we had named as the weak ones were the uniformity in τ of the shifted contour in Lemma 22, where we have given the Stirling estimate in outline rather than in full, and the passage from the spectral expansion of $P_e[\Phi_t^b]$ in L^2 to its pointwise evaluation at the Heegner points, which we justify by the absolute convergence supplied by the Riesz factor and which is the reason the theorem is stated for $m \geq 2$ rather than $m \geq 1$. This paper has twice contained an argument that appealed to a classical theorem about a different object (Remarks 7 and 15); we would rather name the weak points than let a reader hunt for them.

6 Numerical evidence

All computations are exact rational or high-precision arithmetic; the scripts are in the repository ([research/paper-IV/scripts/](#)). For each f and u we compute the piece $\mathcal{P}_u(Y)$ of the set-up paragraph ($= \mathcal{P}_u^{(0)}(Y) + Y/2$ by (3)) for all $Y \leq 10^7$ (a sieve of $Q_u(h)$ for $h \leq Y$, with $\mathbb{E}F_u$ evaluated to 22 significant digits, since an error δ in $\mathbb{E}F_u$ enters as $\delta Y^2/2$), sample it on a logarithmic grid of 4096 points in $\log Y \in [\log 10^3, \log Y_{\max}]$, remove the mean and the linear trend in $\log Y$, and regress the result on

$$\{\cos(t_j \log Y), \sin(t_j \log Y)\}_{j=1}^6$$

for the six smallest even spectral parameters, for the six smallest odd ones, and for 300 random sets of six frequencies drawn uniformly from $[12, 25]$. Reported is the fraction of variance explained, R^2 , and where it falls among the random sets.

Table 2: Fit of $\mathcal{P}_u(Y)/\sqrt{Y}$ to the even and to the odd spectral parameters of $\text{SL}_2(\mathbb{Z})$, against 300 random frequency sets of the same size ($Y \leq 10^7$; $u = 1$ throughout except where shown).

f	D	R^2 even	percentile	R^2 odd	percentile	verdict
$t^2 - 2$	8	0.217	100	0.036	34	even
$t^2 + 1$	−4	0.152	100	0.028	20	even
$t^2 + t + 1$	−3	0.139	100	0.037	35	even
$t^2 + t + 2$	−7	0.089	100	0.051	76	even
$t^2 - 3$	12	0.066	99	0.037	66	even
$t^2 + t - 1$	5	0.062	92	0.036	32	even
$t^2 + t + 5$	−19	0.047	92	0.044	88	inconclusive
$t^2 + 2$	−8	0.031	47	0.054	86	none
$t^2 + t + 3$	−11	0.009	44	0.023	100	none

Table 2 is the evidence for Proposition 9. Six of the nine polynomials place the even set at the 92nd percentile or above, four of them above every one of the 300 random sets; the odd set

never rises above the 88th percentile except once, at the 100th for $t^2 + t + 3$, where the even set is at the 44th and the fit is at the level of the noise. For the dominant line $t_1 = 13.7798$ the fitted amplitude and phase are stable when the range of $\log Y$ is cut in half: for $t^2 - 2$, amplitudes 0.019 and 0.018 with phases 0.89 and 0.61; for $t^2 + t + 1$, 0.021 and 0.017 with phases 1.01 and 1.03; for $t^2 + 1$ at $u = 2$, 0.042 and 0.042 with phases -0.30 and 0.07 . A spurious fit does not reproduce its phase on an independent range.

The periods. Using the Fourier coefficients of the first even form (LMFDB label 1.0.1.3.1, $t_1 = 13.7797513$) we evaluated u_1 at the Heegner points of the nine imaginary discriminants of class number one (`scripts/maass-period.py`).

Table 3: The Katok–Sarnak period of the first even Maass form at the Heegner point of discriminant $D < 0$, class number one, relative to $D = -4$; and the fitted amplitude of the line t_1 , relative to $D = -4$.

D	-3	-4	-7	-8	-11	-19	-43	-67	-163
$ u_1(z_D) / u_1(i) $	0.649	1	0.542	0.318	0.302	0.444	$6.6 \cdot 10^{-3}$	$1.0 \cdot 10^{-4}$	$2.2 \cdot 10^{-10}$
fitted amplitude ratio	0.483	1	0.111	0.281	0.461	1.358	—	—	—
signal present	yes	yes	yes	no	no	marginal	no	no	no

Table 3 confirms the mechanism at the level of orders of magnitude and no further. The threshold (14) sits at $|D| \approx 19.2$: below it every period lies within a factor 3 of every other, and every one of the six discriminants with $|D| \leq 19$ has an amplitude within a factor 12 of the others; above it the periods fall by two, four and ten orders of magnitude at $D = -43, -67, -163$, and no signal is seen at any of them. In particular the absence at $D = -163$, the polynomial $t^2 + t + 41$, is explained. What the table does *not* confirm is the amplitude law itself: the two rows disagree by as much as a factor 5 at $D = -7$. Two unmodelled effects are of that size, the local factors, since $\mathbb{E}F_u$ ranges over $[0.35, 1.60]$ across these polynomials and enters multiplicatively, and the uncertainty of the fitted amplitudes themselves, which for $D = -7$ is a factor 2 between the two halves of the range. We record the amplitude law as a prediction to be tested, not as a confirmed fact.

The weight, and a model piece. The absences at $D = -8$ and $D = -11$, whose periods are of ordinary size, are not explained by anything above, and they are not isolated: among the fourteen polynomials we tested, the six showing no signal are exactly those for which the prime 3 splits in $\mathbb{Q}(\sqrt{D})$. That is a statement about the weight, not about the roots. Indeed $\lambda(3) = 3/(3-4) = -3$ is the only negative value taken by λ and the largest in modulus, and $|\lambda(3)\rho(3)/3| = 2 > 1$, so at $\Re s = \frac{1}{2}$ the local factor at 3 of the Euler product for W_k has modulus up to $3 \cdot 2 \cdot 3^{-1/2} = 3.46$: it is the one local factor that can dominate the spectral term. Removing the weight confirms this. Define the *model piece* by replacing λ by 1 in (1), so that F becomes the restricted divisor function $\sum_{d|n, d \text{ adm}} 1/d$ and $\mathcal{P}_u$ is a Riesz mean of σ_{-1} along Q_u . Repeating the regression:

Every entry of the last four columns is at the 99.3rd percentile or above, every odd set falls below the 29th, and the two discriminants that showed nothing under the weight show a clean signal without it. So the apparent rule at $p = 3$ is an artefact of λ ; nothing arithmetic is hidden there. Note also that the fit improves monotonically along the three columns, for every one of the six discriminants: the more of the arithmetic weight one removes, the more of the variance the even spectrum explains. That is itself evidence that the spectral term is real and that the weight is what obscures it.

Table 4: Removing the arithmetic weight, at $Y \leq 10^7$, $u = 1$. Fit of $\mathcal{P}_u/\sqrt{Y}$ to the six even spectral parameters, for the weighted piece of part III, for the same sum with $\lambda \equiv 1$, and for the sum over all divisors rather than the squarefree ones. The odd sets, omitted, run from the 4.7th to the 28.7th percentile throughout.

D	weighted, squarefree		$\lambda \equiv 1$, squarefree		$\lambda \equiv 1$, all divisors	
	R^2 even	pct	R^2 even	pct	R^2 even	pct
8	0.217	100	0.319	100	0.420	100
-4	0.152	100	0.237	100	0.361	100
-3	0.139	100	0.139	100	0.208	100
-8	0.031	47	0.109	100	0.162	100
-11	0.009	44	0.078	100	0.167	100
17	0.078	99.3	0.098	99.3	0.116	100

The last column is the cleanest object of all. Dropping both the weight and the squarefree condition leaves

$$\mathcal{G}(Y) = \sum_{h \leq Y} (Y - h) (\sigma_{-1}^*(h^2 - D) - \mathbb{E} \sigma_{-1}^*) - \frac{1}{2} (\mathbb{E} \sigma_{-1}^* - 1) Y, \quad \sigma_{-1}^*(n) = \sum_{\substack{d|n \\ (d, 2D)=1}} \frac{1}{d}, \quad (25)$$

a Riesz mean of the divisor-type sums of Hooley and Gafurov. The linear term is the one of the set-up paragraph; with it, $\mathcal{G}(Y) = \int_0^Y S(t) dt$ at integer Y for the sawtooth series S of Step 2 of §5, and it is what the scripts compute. The condition $(d, 2D) = 1$ is retained; it is what makes the levels $e \mid \text{rad}(2D)$ appear (Step 2 of §5 and Remark 13). Lemmas 1 and 2 never used λ or squarefreeness, so they apply with the modulus $d = 1$ added (Step 2), and the Dirichlet series attached to (25) is

$$Z_k(s) = \sum_{(d, 2D)=1} \mathcal{W}_k(D; d) d^{-s},$$

a Salié zeta function carrying a coprimality condition and nothing else. This is the object of §5. The piece the off-diagonal of part III actually requires is (25) perturbed twice: by the squarefree condition, through Möbius over ℓ^2 , and by $\lambda = 1 * \kappa$ with $\kappa(p) = 4/(p - 4) = O(1/p)$.

The phases: a test with no free parameter. The sharpest check available to us is not the size of the oscillation but its phase. In (17) the length enters the test function only through the combination $t/\sqrt{|D|}$, so the spectral term carries the factor $(\sqrt{|D|}/2t)^{it_j}$ of (22), that is $(\sqrt{|D|}/2)^{it_j}$ in its dependence on D ; and by Step 3 the amplitude is a real number, a signed combination of periods of the real-valued form u_j , whose *sign* therefore shifts the phase by π . (The sign used below is that of the Katok–Sarnak period at D , as in the first version of this paper; the corrected Step 3 makes the coefficient a combination over $e \mid \text{rad}(2D)$ in which the level-one term is expected to dominate, and the agreement found here is evidence that it does.) Writing the fitted oscillation as $A_D \cos(t_1 \log Y - \varphi_D)$, the mechanism predicts, once one discriminant is fixed as a reference and with nothing else at our disposal,

$$\varphi_D - \varphi_{D_0} = \frac{t_1}{2} \log \frac{|D|}{|D_0|} - \pi \cdot [\text{sign Per}_D \neq \text{sign Per}_{D_0}] \pmod{2\pi}. \quad (26)$$

The signs of the periods are computed from the Fourier coefficients of the first even form, independently of the arithmetic data. Taking $D_0 = -4$ (`scripts/phase-test.py`):

Every one of the five predictions is correct to within 0.212 radians, that is to 3.4% of a full period. Five independent phases falling that close to prediction by chance has probability

Table 5: The phase of the $t_1 = 13.7798$ oscillation of the model object, predicted by (26) with no free parameter, against the phase fitted to the data ($Y \leq 10^7$, $u = 1$). The reference $D_0 = -4$ is fitted, the other five are predictions.

D	sign Per_D	predicted φ_D	observed φ_D	difference
-3	+	+1.158	+1.050	+0.108
-4	+	+3.140	+3.140	(reference)
-7	+	+0.712	+0.500	+0.212
-8	-	-1.509	-1.430	-0.079
-11	-	+0.685	+0.530	+0.155
-19	-	-1.833	-1.870	+0.037

about $1.4 \cdot 10^{-6}$. The test uses both halves of the mechanism at once: the factor $(2/\sqrt{|D|})^{it_1}$, which is what makes the predicted phases differ from one another at all, and the sign of the Katok–Sarnak period, which is what puts $D = -8, -11, -19$ on the opposite side. We regard this, rather than any of the variance fits, as the evidence that the mechanism of §5 is the right one.

For the amplitudes the agreement is weaker but not absent. Dividing the fitted amplitude by $|\text{Per}_D(u_1)|/|\Gamma_{z_D}|$ gives, in units of 10^7 , the values 2.68, 2.22, 0.39, 1.90, 2.71, 2.71 for $D = -3, -4, -7, -8, -11, -19$: five of the six agree to within 40% with no further factor, and the outlier is $D = -7$, which is also the case whose fitted amplitude is least stable, moving by a factor 1.7 between the two halves of the range.

6.1 The same test on Hooley’s sum

Everything above concerns a divisor sum smoothed by a Riesz mean. The identity (18) predicts the same law for a barer object, namely the partial sums of the Weyl sums themselves,

$$T_k(X) = \sum_{\substack{d \leq X \\ (d, 2D)=1}} \sum_{b \bmod d, b^2 \equiv D} e\left(\frac{kb}{d}\right),$$

the sum Hooley bounded by $O(X^{3/4+\varepsilon})$ in [6]. Since the poles of its Dirichlet series are those of (18), at $s = \frac{1}{2} \pm it_j$, the prediction is $T_k(X) = \sqrt{X} \sum_j c_j X^{it_j} + \dots$, with no smoothing anywhere. We computed $T_1(X)$ for all $X \leq 10^6$ (`scripts/weyl-partial.ts`; roots of $b^2 \equiv D$ by Tonelli–Shanks, Hensel and the Chinese remainder theorem).

$T_1(X)/\sqrt{X}$ is indeed bounded, staying below 0.56 in absolute value with standard deviation 0.14 for $D = -4$: square-root cancellation, well beyond the proved exponent $\frac{3}{4}$. Its spectrum is the even one, for every discriminant tested, at the 96th percentile or above, while the frequencies $2t_j$ are rejected at the 12.7th. And the phase test of (26) applies verbatim, with $(\sqrt{|D|}/2)^{it_1}$ entering through the factor $(\sqrt{|D|}/2t)^{it_1}$ of (22):

Three of the five predictions are correct to within 0.012 radians, two parts in a thousand of a full period, and the median discrepancy over the five is 0.012. We know of no way to obtain agreement of this kind other than by the mechanism of §5 being correct.

The limits of the present computations, and what is planned. Every number reported above comes from a sieve of $Q_u(h)$ for $h \leq 10^7$, which runs in seconds to minutes on a laptop; the scripts are in the repository and each states its cost in its header. Three of the questions raised here are limited by that range rather than by anything conceptual, and are settled by longer runs of exactly the same code:

Table 6: The same parameter-free phase prediction (26), applied to the partial sums of Hooley’s Weyl sums $T_1(X)$, $X \leq 10^6$. Reference $D_0 = -4$; the other five are predictions.

D	R^2 even	percentile	predicted φ_D	observed φ_D	difference
-3	0.173	100	-2.992	-2.980	-0.012
-4	0.412	100	-1.010	-1.010	(reference)
-7	0.159	100	+2.846	+3.110	-0.264
-8	0.252	98.3	+0.624	+0.520	+0.104
-11	0.170	100	+2.818	+2.810	+0.008
-19	0.206	96.0	+0.301	+0.310	-0.009

1. the amplitude law of Table 3, whose comparison is currently blunted by an uncertainty of up to a factor 2 in the fitted amplitudes; the uncertainty falls like $Y^{-1/2}$ in the length of the $\log Y$ range, so $Y = 10^9$ would roughly halve it and $Y = 10^{12}$ would settle the discrepancy at $D = -7$;
2. the resolution in the frequency, which is $2\pi/\log(Y_{\max}/Y_{\min}) \approx 0.68$ at present and therefore cannot separate the lines 19.42 and 19.48, the first even and the first odd parameter of that size; this is the single measurement that would test the parity rule against an adjacent pair rather than against a whole set;
3. the selection rule at $p = 3$, where the sample is fourteen polynomials and should be a few hundred.

None of the three needs a new method, only more arithmetic, and we intend to carry the computation out on appropriate hardware and report it separately. We state them here so that a reader can see exactly which of our assertions rest on 10^7 and which do not: the parity rule of Table 2 and the threshold of Table 3 are stable across the halves of the present range and do not, the amplitude law does.

7 Uniformity in u , and what the Cesàro conjecture would need

Theorem 25 is for one piece, $u = 1$. Part III reduces the Cesàro form of Conjecture 1 of part I to the windows of *all* the pieces with $u \leq H^{2/3+\varepsilon}$, so what is wanted is Theorem 25 uniformly in u , with an error that survives summation against the weight $w(u)$ over that range. This section records what that requires, and why the proof above does not supply it.

For $u > 1$ the roots are those of $u^2x^2 \equiv D$, and by Lemma 5 the dilation moves the frequency from k to $k\bar{u}$, a frequency depending on the modulus. Geometrically, the pairs (d, x) with $u^2x^2 \equiv D$ (d) are again Heegner points, now of discriminant $4u^2D$: the associated form is $[d, 2u^2x, *]$ and the point is

$$z = \frac{-u^2x + u\sqrt{D}}{d}, \quad \Im z = \frac{u\sqrt{|D|}}{d}, \quad \Re z = -\frac{u^2x}{d}.$$

Here is the difficulty, in the sharpest form we can state it. The forms that occur are those of discriminant $4u^2D$ with $2u^2 \mid b$ and $u^2 \mid c$ (and, before the Möbius inversion of Step 2 of §5, $(a, 2Du) = 1$); by the computation of Lemma 11 this set is stable under $\Gamma^0(u^2) = \{\gamma : u^2 \mid \beta\}$, a group conjugate to $\Gamma_0(u^2)$, and under no larger one of the natural candidates,¹³ so that piece

¹³Checked numerically for $(D, u) = (-4, 2), (-8, 3), (-11, 3), (-4, 5), (-3, 2)$ against $\Gamma^0(u^2)$, $\Gamma_0(u^2)$, $\Gamma^0(u)$ and $\Gamma(u)$: stable under the first, not under the other three. Note $z_{[a, 2u^2x, *]} = u z_{[a, 2ux, *]}$ with $[a, 2ux, *]$ of discriminant $4D$: piece u is the level- u sub-family of the discriminant- $4D$ points, dilated by u .

u lives on the surface of level u^2 . Equivalently, since x runs modulo d , the coordinate $\Re z$ runs modulo u^2 , not modulo 1: the seed is invariant under $z \mapsto z + u^2$ but not under $z \mapsto z + 1$, and the stabiliser of ∞ in $\Gamma^0(u^2)$ is exactly the group generated by $z \mapsto z + u^2$. The relevant Poincaré series is therefore taken over $\Gamma_\infty^{(u^2)} \backslash \Gamma^0(u^2)$, and in the pairing with a form of level u^2 at the cusp of width u^2 only the Fourier modes with $k \equiv 0 \pmod{u^2}$ of the level-one forms survive. Consequently the geometric factor is *not* the full Katok–Sarnak period over the Heegner points of discriminant $4u^2D$, but a sum over a restricted sub-family; the relation of Shimura, which would express the full period at Df^2 through Hecke eigenvalues at f and so cost only $u^{7/64+\varepsilon}$, does not apply to it. This is exactly the dilation obstruction of part III, seen geometrically.

What does survive is the analysis of §5.2. Carrying it through with the modes $k = nu^2$ gives $\alpha = 2\pi n$ and $\beta = 2\pi nut/\sqrt{|D|}$, so the expansion parameter of Lemma 22 becomes

$$\frac{\alpha}{\beta} = \frac{\sqrt{|D|}}{ut}, \quad (27)$$

which reduces to the $u = 1$ case and predicts, through the factor $(\alpha/2\beta)^{it_j}$, that the phase of the t_j -oscillation shifts by exactly $-t_j \log u$ relative to $u = 1$. That is a prediction with no free parameter, and it holds: for $u = 2$ and the five discriminants $D = -4, 8, -8, -7, 12$ the differences between predicted and observed phases are 0.002, -0.118 , 0.305, 3.005 and 0.302 radians, so four of the five agree to within 0.31 and the fifth, which has the least stable fit of the six, differs by π , i.e. by a sign in the geometric factor.¹⁴

So the structure extends to every u : the argument of §§5–5.2 applies verbatim to $\Gamma^0(u^2)$ for each fixed u , and what is missing is precisely one thing: the size, in u , of the restricted geometric factor, together with the uniformity in the level of the spectral expansion.¹⁵ Summing $\lambda\omega(u)$ times an error $(H/u)^{1/2}u^A$ over $u \leq H^{2/3}$ is $O(H^{1-\delta})$ exactly when $A < \frac{1}{4}$, so a bound $u^{1/4-\varepsilon}$ for that factor would settle the Cesàro conjecture for quadratics. This is the same wall part III meets from the other side, in the guise of the exponent condition $\theta + 6B < 1$: there, no Weyl-sum bound is uniform enough in the frequency; here, no bound is known for the restricted period. We regard the two as one open problem seen from two directions, but the present formulation is sharper, because the object to be bounded is now a specific sum of a Maass form over a specific set of Heegner points, rather than an error term.

What the present paper does say about it is that the object on the far side is now identified. The windows of part III are not an amorphous error term: piece u carries the spectrum of level

¹⁴Computed from the model object at $Y \leq 10^7$ with the mean value evaluated to 22 digits (`scripts/ef-general.py`); the median discrepancy over the five is 0.302 radians.

¹⁵A first look at $u = 3$ and $u = 5$ (model object, $Y \leq 10^7$, `scripts/piece-divset.ts` with `U`) showed that the level-one lines, which carry the signal at $u = 1$ and $u = 2$, are no longer detectable, and that the object is dominated by slowly varying components, single lines near 3.5 and 5.1 in $\log Y$, which on 13 September we could not identify. They are identified now (14 September). Strömberg [15] proved that $\Gamma_0(9)$ has no genuinely new eigenvalues: its spectrum consists of the level-one forms, the forms of $\Gamma_0(3)$, their twists, and forms coming from the cycloidal group Γ^3 of index three, the last with multiplicity two and in both parities. His table begins 3.5360 (Γ^3), 4.3881 ($\Gamma_0(3)$, odd), 5.0987 ($\Gamma_0(3)$, even), 5.5041 (Γ^3), 6.1206 (odd), 6.6466 (Γ^3), 6.7574 (odd), 7.4318 (Γ^3), 8.0389 (even), 8.6983 (Γ^3). The smooth-window pieces at $u = 3$ for $D = -8, -11, -20$ ($Y \leq 10^7$) have their peaks at 3.48–3.56, 5.08–5.18, 6.59–6.78 and 8.5–8.7, and only traces ($R^2 \leq 0.02$) at the odd-only eigenvalues 4.388, 6.121, 6.757: this is the even part of the $\Gamma_0(9)$ spectrum, as the level- u^2 picture predicts, on a level for which the LMFDB has no data. In the joint-fit test of Table 2 the six even eigenvalues {3.5360, 5.0987, 5.5041, 6.6466, 7.4318, 8.6983} explain 60–82% of the variance of the sharp pieces at $D = -8, -11$ (both objects), at the 95th–100th percentile of 300 random six-sets from [3, 9.5], while the six odd-only eigenvalues explain 8–14%, at the 1st–5th percentile (`research/explore/freq-set-test.ts`). For $u = 5$ the strongest line is at 2.625; Strömberg conjectures that $\Gamma_0(25)$ likewise has no genuinely new forms and that the forms of the cycloidal group Γ^5 appear with multiplicity two, so we predict 2.625 to be the first eigenvalue of Γ^5 , which we have not been able to check. A phase or amplitude test at level 9 needs the Fourier coefficients of the Γ^3 -type forms, which we do not have. Data and peak tables: `research/explore/STATE.md`.

$4u^2$, with amplitudes the periods at discriminant $4u^2D$, and the Cesàro conjecture is the statement that these spectra, summed over u with the weight $w(u)/u$, exhibit enough cancellation. That is a sharper question than the one part III could pose, and it is the one we would attack next.

Related work in the level aspect. For the full sets of Heegner points the level aspect has been treated: Liu, Masri and Young [13] prove that the Heegner points of level q and discriminant D equidistribute as q and D grow, through a hybrid subconvexity bound for the L -functions of Hecke–Maass newforms twisted by theta series, obtained from L^2 -restriction norms of Maass forms on the Heegner points; Humphries and Nordentoft [7] obtain sparse equidistribution of the geometric invariants of real quadratic fields in the level and subgroup aspects by rewriting the Weyl sums as adèlic period integrals attached to Rankin–Selberg L -functions. The sub-families $\{2u^2 \mid b, u^2 \mid c\}$ of this section are natural objects for those methods. Pascadi [14] controls the exceptional spectrum of $\Gamma_0(q)$ on average through a large sieve inequality, which is the input one would need if exceptional eigenvalues at the levels u^2 had to be handled rather than assumed absent.

AI-assisted research: disclosure

This work was carried out with heavy use of large language models, the Anthropic models Claude Fable 5.1 and Claude Opus 5, used interactively by the author in September 2026 as research assistants. The author posed the questions, set the direction and the scope, and made every decision about what to pursue, claim and submit. The model was used to draft mathematical arguments, to write and run the software behind every number in the paper, to search the literature, and to draft the text; all of this was done in dialogue with the author, who verified the code and the numerical results, reviewed the proofs, and takes full responsibility for the content. The model is not an author. The complete record of the work is in the public repository <https://github.com/gewure/ulamnd> (directory `research/`).

References

- [1] V. A. Bykovskiĭ. Spectral expansions of certain automorphic functions and their number-theoretic applications. *Zap. Nauchn. Sem. LOMI*, 134:15–33, 1984. English transl.: J. Soviet Math. 36 (1987), 8–21.
- [2] W. Duke, J. B. Friedlander, and H. Iwaniec. Equidistribution of roots of a quadratic congruence to prime moduli. *Annals of Mathematics*, 141:423–441, 1995.
- [3] W. Duke, J. B. Friedlander, and H. Iwaniec. Weyl sums for quadratic roots. *Int. Math. Res. Not. IMRN*, (11):2493–2549, 2012.
- [4] Dorian Goldfeld and Peter Sarnak. Sums of Kloosterman sums. *Inventiones Mathematicae*, 71:243–250, 1983.
- [5] Jeffrey Hoffstein and Paul Lockhart. Coefficients of Maass forms and the Siegel zero. *Annals of Mathematics*, 140:161–181, 1994. With an appendix by D. Goldfeld, J. Hoffstein and D. Lieman.
- [6] C. Hooley. On the number of divisors of quadratic polynomials. *Acta Mathematica*, 110:97–114, 1963.

- [7] Peter Humphries and Asbjørn Christian Nordentoft. Sparse equidistribution of geometric invariants of real quadratic fields. *J. Eur. Math. Soc.*, 2025. to appear; arXiv:2211.05890.
- [8] Henryk Iwaniec. *Topics in Classical Automorphic Forms*, volume 17 of *Graduate Studies in Mathematics*. American Mathematical Society, 1997.
- [9] Henryk Iwaniec. *Spectral Methods of Automorphic Forms*, volume 53 of *Graduate Studies in Mathematics*. American Mathematical Society, 2nd edition, 2002.
- [10] Henryk Iwaniec, Wenzhi Luo, and Peter Sarnak. Low lying zeros of families of L -functions. *Publ. Math. Inst. Hautes Études Sci.*, 91:55–131, 2000.
- [11] Svetlana Katok and Peter Sarnak. Heegner points, cycles and Maass forms. *Israel Journal of Mathematics*, 84:193–227, 1993.
- [12] Henry H. Kim. Functoriality for the exterior square of GL_4 and the symmetric fourth of GL_2 . *Journal of the American Mathematical Society*, 16:139–183, 2003. With appendices by D. Ramakrishnan and by H. Kim and P. Sarnak.
- [13] Sheng-Chi Liu, Riad Masri, and Matthew P. Young. Subconvexity and equidistribution of Heegner points in the level aspect. *Compos. Math.*, 149:1150–1174, 2013. arXiv:1206.3208.
- [14] Alexandru Pascadi. Large sieve inequalities for exceptional Maass forms and the greatest prime factor of $n^2 + 1$. *Forum Math. Pi*, 14:e8, 2026. arXiv:2404.04239.
- [15] Fredrik Strömberg. Newforms and spectral multiplicities for $\Gamma_0(9)$. *Proc. Lond. Math. Soc.*, 2012. arXiv:1106.5741.
- [16] Kazunari Sugiyama. Shintani correspondence for Maass forms of level N and prehomogeneous zeta functions, 2021. arXiv:2110.02847; announcement, details to appear.
- [17] Matthew Welsh. Parameterizing roots of polynomial congruences. *Algebra & Number Theory*, 16:881–918, 2022.